The galaxy-bias ladder across thin redshift bins

A correlated linear-bias prior, the one-loop fiducial model, and the reach of DESI C_ℓ^{gg}

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1 Summary

The question. Thin redshift bins measure growth only if each bin's galaxy bias is tied to its neighbors. For DESI angular band powers C_ℓ^{gg} alone on four bin layouts (Section 3), this note sets how freely a thin bin's linear bias may move within its tracer group, fits the one-loop fiducial bias model of `notes/higher_order_bias` with and without its tilts (here with α_2 free and the exploration σ_v widths), tests which choices recover the cosmology honestly, and gives the figure of merit (FoM) against the scale cut $k_{\rm max}$.

The linear-bias prior. Each bin's bias follows its group's growth-tied trend B_g/D times a free straight line in redshift, plus a residual with a correlated Gaussian prior of width σ_g and correlation length L (Section 4). Four priors span the range: growth-tied (no line, no residual), baseline ($\sigma_g = 3\%$ for BGS and 1% elsewhere, $L = 0.2$), pessimistic (3% everywhere, $L = 0.05$) and free (one free bias per bin).

The bias model. The fiducial model adds to each bin's linear bias the one-loop terms: quadratic and tidal biases on calibrations plus one offset per group, the counterterms α_0 and α_2, and the stochastic terms SN_0 and $\mathrm{SN}_2\, k^2\mu^2$ (Section 5). Every per-group quantity carries the same structure as the linear bias: a group value, a straight line in redshift (free when the tilts are fitted), and, for b_1 alone, a correlated per-bin residual.

The accuracy test. At every cut, the fiducial model fits truths that carry the known drifts plus a residual drawn from the pessimistic prior, and each fit is scored by $\sqrt{\langle\chi^2\rangle/n_{\rm dof}}$, a statistic that equals 1 when the quoted error bars cover the truth as often as they claim (Section 6). On the pessimistic prior its error bars stay honest, $\sqrt{\langle\chi^2\rangle/n_{\rm dof}} \le 1.03$ in every score. On the growth-tied prior they collapse: the statistic reaches 3.9, 4.5 and 3.1 on $\Delta d_{\rm com} = 62$ Mpc, $\Delta z = 0.02$ and 0.05, and at $k_{\rm max} = 0.2\,h$/Mpc the Ω_m error bar covers the truth in 2.5–10% of universes instead of 68%. The growth-tied prior is discarded and gets no FoM.

The fiducial model against $k_{\rm max}$ and the result. Two scenarios pair the baseline prior with the fiducial model without tilts and the pessimistic prior with the fiducial model, fitted against $k_{\rm max}$ to truths that carry α_2, SN_2, the offsets at one prior width, or the calibration drift; a truth inside the fitted model's own prior is reported, one outside it is held to $\sigma/3$ (Section 7). The baseline scenario breaks at $k_{\rm max} = 0.075\,h$/Mpc on the thin layouts and at 0.05 on the wide ones, through the calibration drift; from 0.1 on its largest tested shift is 0.53–2.8σ. The pessimistic scenario absorbs the drift exactly and passes every cut; its reported in-prior shifts reach 0.98σ at 0.075 on $\Delta d_{\rm com} = 62$ Mpc and 1.10σ at 0.1 on $\Delta z = 0.02$, mostly through α_2, and 0.33–0.73σ from 0.125 onward on the three thinner layouts. These shifts estimate how safely an MCMC with the model would run on real data. The fiducial model

parameter	meaning	fiducial
$10^9 A_s$	amplitude of the primordial curvature power	2.1
n_s	scalar spectral index	0.96605
H_0	Hubble constant [km/s/Mpc]	67.32
Ω_b	baryon density today	0.04
Ω_m	matter density today (neutrinos included)	0.3
w_0	dark-energy equation of state today	-1
w_a	its slope, $w(a) = w_0 + w_a(1-a)$	0
$\sum m_\nu$	neutrino mass sum [eV], fixed	0.06
τ	optical depth to reionization, fixed	0.0697186

Table 1: The fiducial cosmology. The first seven parameters are varied in every Fisher matrix, with flat priors.

on the pessimistic prior gives FoM$(w_0, w_a) = 0.407$, 0.237, 0.155 and 0.0426 at 0.075 and 7.53, 8.42, 2.45 and 0.300 at 0.2 (layout order); the baseline scenario gives 1.12 and 1.18 times its (w_0, w_a) FoM on the thin layouts at 0.2. Both scenarios stay below LSST Y1 3x2pt in both FoMs. Appendices P and Q give the limits and the open issues.

2 Fiducial model and parameter values

This section freezes the fiducial model and every parameter value the note's figures were computed with, so the memory does not depend on configuration files that drift over time.

Cosmology. Flat $w_0 w_a$CDM. The Fisher matrices vary the seven parameters of Table 1 with flat priors; the neutrino mass sum is fixed at 0.06 eV and the optical depth at $\tau = 0.0697186$. w_a is sampled as $w_0 + w_a$. At this point $\sigma_8 = 0.8278$.

Layouts. Four DESI bin layouts, each with the five LSST Y1 source bins and 53 band powers per spectrum (averages of C_ℓ over multipole bands of width $\Delta\ell = 10$, $\ell = 15$ to 544): bins of equal comoving width $\Delta d_{\rm com} = 62$ Mpc (76 lens bins, 107 cross pairs, 10,494 entries), and bins of fixed width $\Delta z = 0.02$ (90 bins, 133 pairs, 12,614 entries), 0.05 (36 bins, 52 pairs, 5,459 entries) and 0.2 (10 bins, 13 pairs, 2,014 entries). The six tracer groups are BGS, LRG1, LRG2, LRG3, ELG1 and ELG2.

Linear bias. Bin i of group g has $b_{1,i} = B_g(1+\delta_i)/D(\bar{z}_i)$, with D read from the linear power spectrum ratio at $k = 0.005\,\mathrm{Mpc}^{-1}$ and every deviation $\delta_i = 0$ at the fiducial, so $b_{1,i} = B_g/D(\bar{z}_i)$. The fiducial amplitudes are $B_g = 1.34$ (BGS), 1.7 (LRG1, LRG2, LRG3) and 0.84 (ELG1, ELG2). The correlated priors of the forecast are the stated widths of Section 4: baseline $\sigma_g = 3\%$ for BGS and 1% elsewhere with $L = 0.2$; pessimistic $\sigma_g = 3\%$ in every group with $L = 0.05$.

Higher-order terms. $b_{2,i} = b_2^{\rm L}(b_{1,i}) + \Delta_{2,g}$ and $b_{s^2,i} = -\frac{4}{7}(b_{1,i}-1) + \Delta_{s,g}$ (Eq. (5)); $b_3 = 0$. Every offset, tilt, counterterm and stochastic amplitude is zero at the fiducial, so b_2 and b_{s^2} sit on their calibrations. The prior widths are the fiducial model's (Section 5): 2 on Δ_2, 1 on

	BGS0	BGS1	LRG1	LRG2	LRG3	ELG1	ELG2
fiducial [km/s]	371	358	357	348	343	304	289
prior width [km/s]	145	105	70	50	50	40	30

Table 2: The velocity-dispersion fiducials and Gaussian prior widths of the seven σ_v groups (BGS0: $\bar{z}_i < 0.255$).

Δ_s, 2 on SN_0, 5 on SN_2, and $0.5/k_{\mathrm{max}}^2$ on α_0 and α_2 at each cut of the scan. The SN_2 unit $f_{\mathrm{sat}}\sigma_{\mathrm{sat}}^2$ is $6.084\,(\mathrm{Mpc}/h)^2$ for BGS and LRG and $0.841\,(\mathrm{Mpc}/h)^2$ for ELG.

Velocity dispersions. The Fingers-of-God smearing (Section 5) convolves each bin's redshift distribution with a Gaussian of width $(1+\bar{z}_i)\,\sigma_v/c$, with one sampled σ_v per group and BGS split at $\bar{z} = 0.255$ (Table 2). The yamls truncate each prior at $\sigma_v = 0$; the Fisher matrices treat it as an untruncated Gaussian, which at these widths changes only the BGS0 mean, by 2 km/s. The paper configuration of the Fisher notebook widens every width by a factor 6, at which the BGS0 truncation matters (the truncated mean is about 850 km/s); this note's fits use the widths of Table 2.

Scale cuts. The UV-fraction cuts (Section 3) are computed on the paper model's fiducial spectra, the model above at the evaluate point of `EXAMPLE_EVALUATE1`–`4.yaml`, with $\alpha = 0.05$ and $k_{\mathrm{max}} = 0.05$ to $0.25\,h/\mathrm{Mpc}$ in steps of 0.025 (the scan uses at most 0.2). They were computed at the final derivative tier, the settings of the derivative base (the Derivatives paragraph below), and cached in `chains/model_scale_cuts_final.npz`. The shipped mask stops every bin at $k = 0.1185\,\mathrm{Mpc}^{-1}$.

Covariance. The Gaussian term of the production archives, computed for 14,000 square degrees with the inverse rescaled by $12300/14000 = 0.879$, at the one-loop fiducial above with one difference: massless neutrinos ($\sum m_\nu = 0$). Its non-Limber χ grid runs at twice the likelihood's base accuracy. The LSST Y1 3x2pt reference values, $\mathrm{FoM}(w_0, w_a) = 17.26$ and $\mathrm{FoM}(\Omega_m, 10^9 A_s) = 214.9$, come from the `lsst_y1` project's real-space SRD Fisher forecast (1560 data points, steps $h = 0.025$), whose fiducial differs from this note's: $w_0 = -0.9$, and nonzero photo-z and shear-calibration fiducials under their Gaussian priors.

Derivatives. Every stencil derivative (the cosmology, the amplitudes B_g, the dispersions σ_v) is a five-point stencil with offsets $(2, 1, -1, -2)$ and relative step $h = 0.02$; the per-bin operator columns use steps of 1% of b_1, 0.02 in b_2 and b_{s^2} and $5\,h^{-2}\,\mathrm{Mpc}^2$ in α_0; α_2, SN_0 and SN_2 are exact linear templates. All run at the final derivative tier of the convergence study: CosmoLike accuracy boost 3, non-Limber boost 64, integration accuracy 1, Boltzmann k range $20\,\mathrm{Mpc}^{-1}$, and CAMB with accuracy boost 2, 30 wavenumbers per decade and the halofit tolerance patch.

Theory figures. The drift laws and prior draws (`prior_draws`, `theory_*`) use flat ΛCDM with $\Omega_m = 0.3$ and no scale cut.

3 The forecast

The data are the DESI angular band powers C_ℓ^{gg} of thin redshift bins: the angular power spectrum of galaxy clustering, averaged in bands of multipole ℓ, and computed without the Limber approximation, which ties each multipole to a single wavenumber at the bin's distance and fails for thin bins. The spectra cover six tracer groups (BGS, LRG1, LRG2, LRG3, ELG1 and ELG2) on four bin layouts: bins of equal comoving width $\Delta d_{\rm com} = 62$ Mpc, and bins of fixed redshift width $\Delta z = 0.02$, 0.05 and 0.2. Bin i has mean redshift $\bar{z}_i$, and $D(z)$ is the linear growth factor, $D(0) = 1$. $w(a) = w_0 + w_a(1 - a)$ is the dark-energy equation of state, with a the scale factor; Ω_m is the matter density today in units of the critical density, and A_s the amplitude of the primordial curvature power spectrum, sampled as $10^9 A_s$. A parameter's marginalized one-sigma error is σ: the error after every other parameter is fitted as well, with their uncertainty included. $\mathrm{FoM}(w_0, w_a) = [\det \mathrm{Cov}(w_0, w_a)]^{-1/2}$ is the inverse area of the marginalized (w_0, w_a) error ellipse, and $\mathrm{FoM}(\Omega_m, 10^9 A_s)$ is defined the same way. The reference is LSST Y1 3x2pt alone: the `lsst_y1` project's real-space SRD Fisher forecast [7], computed on a different data vector and at a slightly different fiducial (Section 2).

The scale cut. Bin i is cut at $k_{\rm max}/D(\bar{z}_i)$: matter is less nonlinear at higher redshift, so the cut moves to higher wavenumber k as $1/D$, and every $k_{\rm max}$ in this note is the $z = 0$ value. The cut is the non-Limber UV-fraction cut of Saraivanov et al. [1], which follows DeRose and Chen [2]. Each band power is split into $C_\ell^{\rm IR}$, from the power spectrum below the bin's cut wavenumber, and $C_\ell^{\rm UV}$, from the power above it, and a band is kept while

$$\frac{|C_\ell^{\rm UV}|}{|C_\ell^{\rm UV}| + |C_\ell^{\rm IR}|} \; < \; 0.05. \tag{1}$$

The kept bands are intersected with the shipped mask, which stops every bin at $k = 0.1185\,\mathrm{Mpc}^{-1}$. The cut is computed on the paper model's fiducial spectra (b_2 and b_{s^2} on their calibrations, every other higher-order term zero; Section 2). The scan runs over $k_{\rm max} = 0.05, 0.075, \ldots, 0.2\,h/\mathrm{Mpc}$.

The covariance. The band-power covariance is Gaussian: it keeps the term in which the band powers scatter as Gaussian fields and drops the connected higher-order terms. It is computed for 14,000 square degrees, and every Fisher matrix rescales its inverse by 12,300/14,000 = 0.879 to the survey area, because the scatter of every band power shrinks as more sky is averaged. It is evaluated at the one-loop fiducial (Section 5): $b_{1,i} = B_g/D(\bar{z}_i)$, b_2 and b_{s^2} on their calibrations, every other term zero. The model shares this fiducial point with every one-loop variant of `notes/higher_order_bias`, so one covariance serves them all. Its one-loop terms come from CosmoLike's C-FAST-PT tables, through the data vector's pair formula, in the Limber approximation; the non-Limber recompute of its lowest bands uses linear theory only (Appendix Q). Its non-Limber grid converges the adjacent thin-bin cross spectra to 2% and the diagonal to 0.4%. It takes massless neutrinos, so its signal sits 3% above the data vector ($m_\nu = 0.06$ eV) at every redshift.

4 The linear-bias prior

The default model ties all bins of a group to one amplitude, $b_{1,i} = B_g/D(\bar{z}_i)$, with $b_{1,i}$ the linear bias of bin i and B_g one free amplitude per group g: a constant clustering amplitude $b_1 D$, as measured for DESI-like luminous red galaxies [14]. The tying makes thin bins powerful, but known physics moves each bin away from the group trend at the percent level (Appendix B), and a drift the model cannot express leaves its quoted error bars covering the truth in a small fraction of universes (Section 6). One free bias per bin is accurate but spends data on the biases, 76 of them on $\Delta d_{\rm com} = 62$ Mpc. The prior below sits between the two: bins may leave the group trend, but neighbors move together.

The linear bias of bin i in group g is

$$b_{1,i} \;=\; \frac{B_g}{D(\bar{z}_i)}\,\big(1+\delta_i\big), \qquad \delta_i \;=\; s_g\,(\bar{z}_i - \bar{z}_g) + \varepsilon_i, \tag{2}$$

with B_g a free amplitude and s_g a free tilt per group (flat priors), $\bar{z}_g$ the mean of the group's bin redshifts, δ_i the fractional deviation from the group trend, and ε_i the residual around the group's line. The residuals carry a Gaussian prior correlated within a group and zero across groups,

$$\big\langle \varepsilon_i\,\varepsilon_j \big\rangle \;=\; \sigma_g^2\,\exp\!\left(-\,\frac{|\bar{z}_i - \bar{z}_j|}{L}\right), \tag{3}$$

with σ_g the prior width of one bin's residual and L the correlation length in redshift. σ_g sets how far one bin may wander from its group's line, L how far apart two bins must be before their residuals are independent, and together they set the rms jump between neighbors at separation Δz, $\sigma_g[2(1-e^{-\Delta z/L})]^{1/2}$ (Appendix A.2).

The tilt is free because B_g/D is not the evolution of a conserved population, whose bias obeys $b-1 \propto 1/D$ [9] (Appendix B.1), and every derived drift is close to a straight line across its group (Appendix C): the line absorbs coherent evolution, and the residual the curvature of a drift and local features of the selection (Appendix C.4). In the Fisher matrix the prior is one projected precision block per group, an inverse-covariance penalty with the level and the line projected out, so that it charges only the residual (Appendix C.5); a line fits one or two bins exactly, so such a group is free. Every group of the $\Delta z = 0.2$ layout is of this kind, and there the baseline, pessimistic and free priors coincide.

The four priors.

- **Growth-tied**: $b_{1,i} = B_g/D(\bar{z}_i)$, with no tilt and no residual; the default of the shipped yamls.
- **The baseline prior**: $\sigma_g = 3\%$ for BGS, 1% elsewhere, and $L = 0.2$. The BGS width holds the 2.4% rms curvature that a straight line leaves of the 22% drift of its $r < 19.5$ flux limit (Appendix E); the 1% is a stated hedge for unmodeled selection curvature, at least six times the largest rms (0.16%) a conserved population leaves after its line (Appendix C.4). $L = 0.2$, comparable to the group spans, keeps the residual smooth across a group.
- **The pessimistic prior**: $\sigma_g = 3\%$ in every group and $L = 0.05$: local selection features as large as the largest derived curvature, on the redshift scale over which the 4000 Å

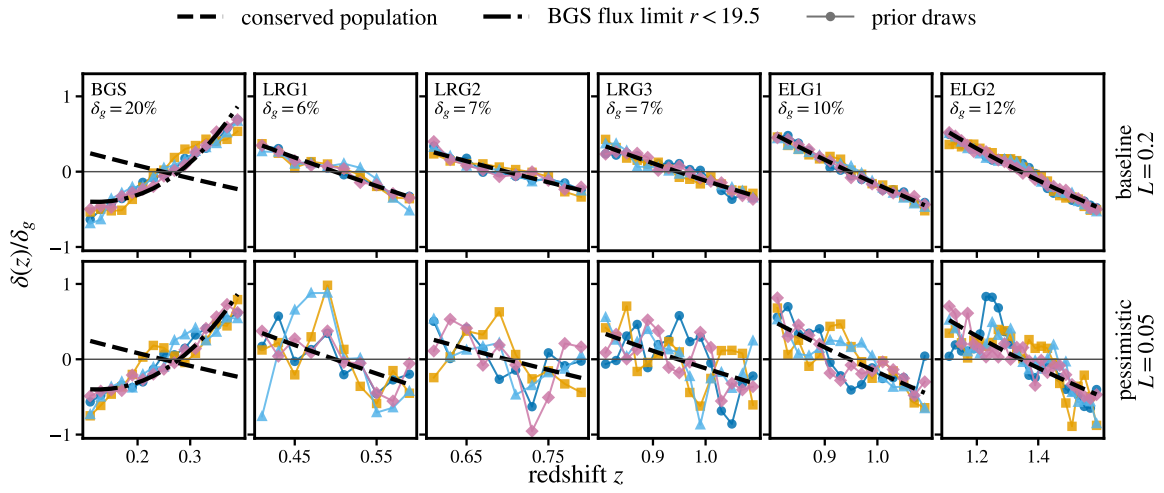


Figure 1: Draws of the baseline prior (top row) and the pessimistic prior (bottom row) on the $\Delta z = 0.02$ bins, one column per group. The vertical axis is $\delta(z)/\delta_g$, with δ_g the round per-group scale printed in the top panel, so every column fills its frame. Thick dashed: the conserved-population drift; thick dash-dot (BGS): the $r < 19.5$ flux-limit drift. Thin curves: four draws of $\delta(z)$ on the line of the group's reference drift, so draw and drift differ by the residual ε.

break crosses a filter edge. That crossing is known, but its amplitude in bias is not measured (Appendix B.3).

- **Free**: one $b_{1,i}$ per bin, each with a flat prior; the lower limit.

Figure 1 draws both priors around the known drifts. Under the baseline prior the neighbor rms of the residual at $\Delta z = 0.02$ is 0.44% (1.31% for BGS), and the known drifts sit well inside the smooth family; under the pessimistic prior it is 2.44% in every group: the histories keep their lines but gain local features nobody has measured.

5 The bias model

Linear bias alone is not enough beyond $k_{\rm max} = 0.05\,h/{\rm Mpc}$: a linear-bias fit to a one-loop truth is biased at 0.075 on every layout and under every prior (`notes/linear_bias_kmax`). The forecast therefore uses the one-loop fiducial bias model of `notes/higher_order_bias`, which defines it in full and validates every term; this section states it briefly. The model keeps the per-bin linear bias of Eq. (2), and the real-space galaxy power spectrum of bin i is

$$P_{gg,i}(k) \;=\; b_{1,i}^2\,P_{\rm mm}(k) \;+\; P_{\rm loop}(k;\, b_{1,i}, b_{2,i}, b_{s^2,i}) \;+\; \alpha_0\,k^2 P_{\rm mm}(k) \;+\; \frac{{\rm SN}_0}{\bar{n}_i}, \qquad (4)$$

where $P_{\rm mm}$ is the nonlinear matter power spectrum and $P_{\rm loop}$ holds the one-loop terms: the first corrections of cosmological perturbation theory beyond linear, built from products of two power spectra. These carry the quadratic bias b_2, the response of the galaxy density to the square of the matter density contrast δ_m (the galaxy density contrast contains $b_2\,\delta_m^2/2$), and

the tidal bias b_{s^2}, its response to the squared tidal field; the third-order bias b_3 is zero. α_0 is the k^2 counterterm, in $h^{-2}\,\mathrm{Mpc}^2$, which absorbs the sensitivity of the loops to small scales, and SN_0 the white-noise power beyond Poisson, in units of the bin's Poisson power $1/\bar{n}_i$, with $\bar{n}_i$ the bin's number density: counting a finite number of galaxies adds the flat shot-noise power $1/\bar{n}_i$, and SN_0 measures the excess over it. The higher-order biases follow each bin's own linear bias through calibrations, plus one offset per group:

$$b_{2,i} \;=\; b_2^{\mathrm{L}}(b_{1,i}) + \Delta_{2,g}, \qquad b_{s^2,i} \;=\; -\tfrac{4}{7}\,(b_{1,i}-1) + \Delta_{s,g}, \tag{5}$$

with b_2^{L} the separate-universe fit of Lazeyras et al. [3] and the second relation the co-evolution of a population formed with local bias [4].

Redshift space. A galaxy is placed at the distance its redshift implies, and its peculiar velocity along the line of sight shifts that redshift, so the inferred map is distorted. The model treats these redshift-space distortions to linear order (the Kaiser effect: coherent infall toward overdensities), and the random motions within groups and clusters, the Fingers of God, by convolving each bin's redshift distribution with a Gaussian set by the sampled velocity dispersion σ_v, one per group (Section 2). Two more terms depend on μ, the cosine of the angle to the line of sight. The counterterm α_2 adds, in the convention of the DESI DR1 full-shape analysis [6],

$$P_{\mathrm{CT}} \;=\; (b_1 + f\mu^2)\, f\, \alpha_2\, \mu^2 k^2 P_{\mathrm{lin}}, \tag{6}$$

with f the linear growth rate and P_{lin} the linear matter power spectrum. The stochastic power $\mathrm{SN}_2\, k^2\mu^2$ is counted in units of $f_{\mathrm{sat}}\sigma_{\mathrm{sat}}^2/\bar{n}_i$, with f_{sat} the satellite fraction and σ_{sat} the satellites' velocity dispersion in distance units: $(0.1, 7.8\,h^{-1}\mathrm{Mpc})$ for LRG, $(0.1, 2.9\,h^{-1}\mathrm{Mpc})$ for ELG, and the LRG values for BGS. Both stochastic terms project exactly, without the Limber approximation:

$$C_\ell^{ij}\big|_{\mathrm{SN}_0} = \mathrm{SN}_0\, P_{ij} \int d\chi\, \frac{W_i\, W_j}{\chi^2}, \qquad C_\ell^{ij}\big|_{\mathrm{SN}_2} = \mathrm{SN}_2\, f_{\mathrm{sat}}\sigma_{\mathrm{sat}}^2\, P_{ij} \int d\chi\, \frac{W_i'\, W_j'}{\chi^2}, \tag{7}$$

with χ the comoving distance, $W_i(\chi)$ the Fingers-of-God-convolved kernel of bin i, $W_i' = dW_i/d\chi$, and P_{ij} the Poisson power of the pair (Appendix M.1). Because $k^2\mu^2$ is the squared line-of-sight wavenumber, SN_2 is white noise differentiated along the line of sight. A thin bin's kernel derivative peaks at its two edges, with opposite signs, and adjacent bins share an edge, so the projected SN_2 is anticorrelated between neighbors: normalized by the two autos, its cross term has a median of -0.50 over the 73 adjacent pairs of one tracer sample on $\Delta d_{\mathrm{com}} = 62$ Mpc, against $+0.05$ for SN_0. No white noise, per group or per bin, can mimic it, and without SN_2 a one-loop fit breaks before its own cut on the thin layouts (`notes/higher_order_bias`).

One structure for every quantity. The model is never defined through b_1 alone. Every per-bin quantity X_i of bin i in group g carries the structure the linear bias has in Eq. (2): a group value (riding its calibration trend where one exists), a straight line in redshift, and a per-bin residual,

$$X_i \;=\; \bar{X}_g(b_{1,i}) \;+\; t_{X,g}\,(\bar{z}_i - \bar{z}_g) \;+\; \varepsilon_{X,i}, \tag{8}$$

with $t_{X,g}$ the tilt of X in group g. Table 3 lists what each quantity carries. **Without tilts**, every tilt is fixed to zero: one Δ_2, Δ_s, α_0, α_2, SN_0 and SN_2 per group, 36 parameters. **With**

quantity	group value	tilt $t_{X,g}$	residual $\varepsilon_{X,i}$
b_1 (as $\beta_i = b_{1,i}D$)	free B_g	free (s_g)	correlated Gaussian (σ_g, L)
b_2, b_{s^2}	calibration + offset (widths 2, 1)	free in + tilts	only through b_1's residual
α_0, α_2	Gaussian, width $0.5/k^2_{\rm max}$	free in + tilts	none
SN_0, SN_2	Gaussian, widths 2 and 5	never tilts	none

Table 3: What each quantity of Eq. (8) carries. The b_2 and b_{s^2} group values are the calibrations of Eq. (5) plus one Gaussian offset per group. Every tilt is fixed to zero in the fit without tilts and free (flat prior) in the fit with tilts, except for SN_0 and SN_2, which never tilt.

tilts, the tilts of Δ_2, Δ_s, α_0 and α_2 are free, with flat priors, in every group of more than one bin: up to 60 parameters. SN_0 and SN_2 never tilt: their Poisson part is known bin by bin. The Gaussian priors are centered on zero, with the DESI DR1 stochastic widths [6] and the counterterm rule of the DESI 2024 full-shape analysis [5, 6] moved to each cut: the width obeys $\alpha\, k^2_{\rm max} = 0.5$, so it is $0.5/k^2_{\rm max}$, which at $k_{\rm max} = 0.2\,h/\mathrm{Mpc}$ gives $0.5/0.04 = 12.5\,h^{-2}\,\mathrm{Mpc}^2$, DESI's value. Since α_0 multiplies $P_{\rm mm}$, at bin i's own cut $k = k_{\rm max}/D(\bar{z}_i)$ the width contributes $\alpha_0 k^2 = 0.5/D^2(\bar{z}_i)$, while the bin's linear galaxy power carries $b^2_{1,i} = B^2_g/D^2(\bar{z}_i)$: the ratio, a correction of $0.5/B^2_g$, is the same in every bin of a group. Appendix M gives the implementation.

6 The accuracy test

No mock catalogs (simulated galaxy surveys against which a prior could be checked) and only two derived drift laws are available, so this note does not defend one prior: it tests each prior against random truths and quotes the FoM only for the priors that pass.

Random truths. Each truth sets every bin's linear bias to its group's known drift (the $r < 19.5$ flux limit in BGS, a conserved galaxy population elsewhere; Appendix K.2) plus a residual drawn from the pessimistic prior, with b_2 and b_{s^2} following b_1 through Eq. (5). The truths are fitted with the fiducial model, centered on the growth-tied shape, under the growth-tied prior and under the pessimistic prior, to first order in the truth; the average over the truths and the data noise is exact (Appendix K).

The statistic. Each fit is scored by

$$\sqrt{\langle \chi^2 \rangle / n_{\rm dof}}, \qquad \chi^2 = \Delta p^{\mathsf{T}}\, \mathsf{C}_p^{-1}\, \Delta p, \tag{9}$$

with Δp the recovered minus the true parameters, C_p the fit's marginalized covariance, $n_{\rm dof} = 2$ for $p = (w_0, w_a)$ and 1 for Ω_m or A_s, and $\langle\cdot\rangle$ the exact average over the data noise and the bias history. It measures how well each fit's quoted error bar covers where the fit actually lands. Its value, written η, is 1 for error bars that cover the truth as often as they claim; above 1 they are too small by that factor, and below 1 conservative. It scores the parameters, not the χ^2 of the data, and for one parameter it converts into the fraction of universes in which the 1σ bar covers the truth (Appendix K.4).

The result. Figure 2 draws the test at every cut. The pessimistic prior keeps $\eta \leq 1.03$ in every score, at every cut and on every layout: fitted to its own scatter, its error bars are honest

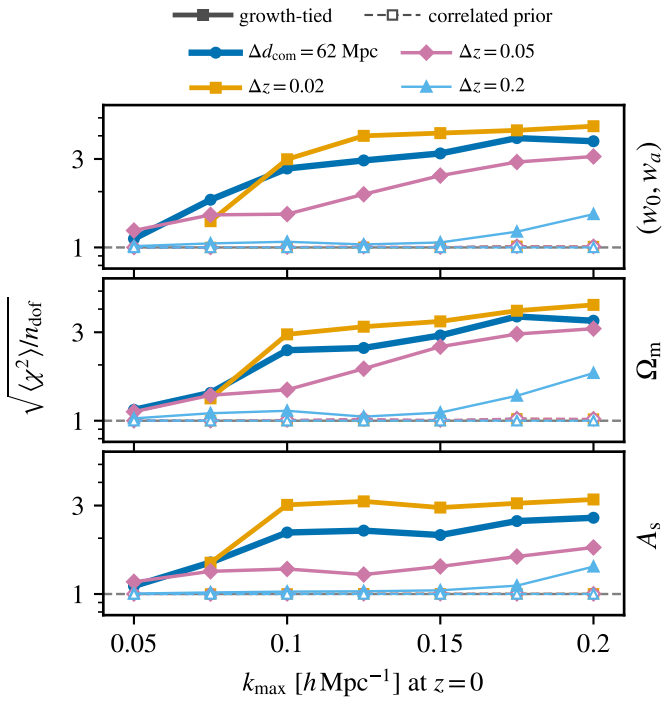


Figure 2: The accuracy test at every cut: $\eta = \sqrt{\langle\chi^2\rangle/n_{\rm dof}}$ of (w_0, w_a), Ω_m and A_s (one panel each, one shared logarithmic scale) for the fiducial model fitted with the growth-tied prior (solid, filled markers) and with the correlated prior, here the pessimistic prior (thin dashed, open markers), to truths that carry the known drifts plus a residual drawn from the pessimistic prior. Color, line width and marker: the layouts. Gray dashed: $\eta = 1$, honest error bars. Omitted: the free prior, which gives exactly 1. $k_{\rm max}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

(Appendix K.4, the matched case). Under the growth-tied prior η grows from 1.0–1.4 at the smallest fitted cut to largest scores of 3.9, 4.5 and 3.1 on $\Delta d_{\rm com} = 62$ Mpc, $\Delta z = 0.02$ and 0.05, and 1.8 on $\Delta z = 0.2$; at $k_{\rm max} = 0.2\,h/$Mpc its Ω_m bar covers the truth in 6.3, 2.5, 10 and 35% of universes instead of 68%. The growth-tied prior is therefore discarded and gets no FoM in any figure. The free prior absorbs any truth of the linear bias and gives exactly 1 (Appendix K.4).

7 The fiducial model against $k_{\rm max}$

The scenarios and the truths. Two scenarios are carried side by side with equal weight: the baseline prior with the fiducial model without tilts, and the pessimistic prior with the fiducial model. Each fit is given a truth that carries one term at one prior width, the same in every bin, or a drift across each group: α_2 at its own prior width, $\pm 0.5/k_{\rm max}^2$; $\mathrm{SN}_2 = \pm 5$; the offsets (Δ_2, Δ_s) at one prior width, $\pm(2, 1)$; or the calibration drift, which moves Δ_2 and Δ_s linearly across each group, by 1 and 0.5 from its first bin to its last with zero mean (Appendix M.4). The shift of the fit is the linear response of Appendix K.1, exact for α_2 and

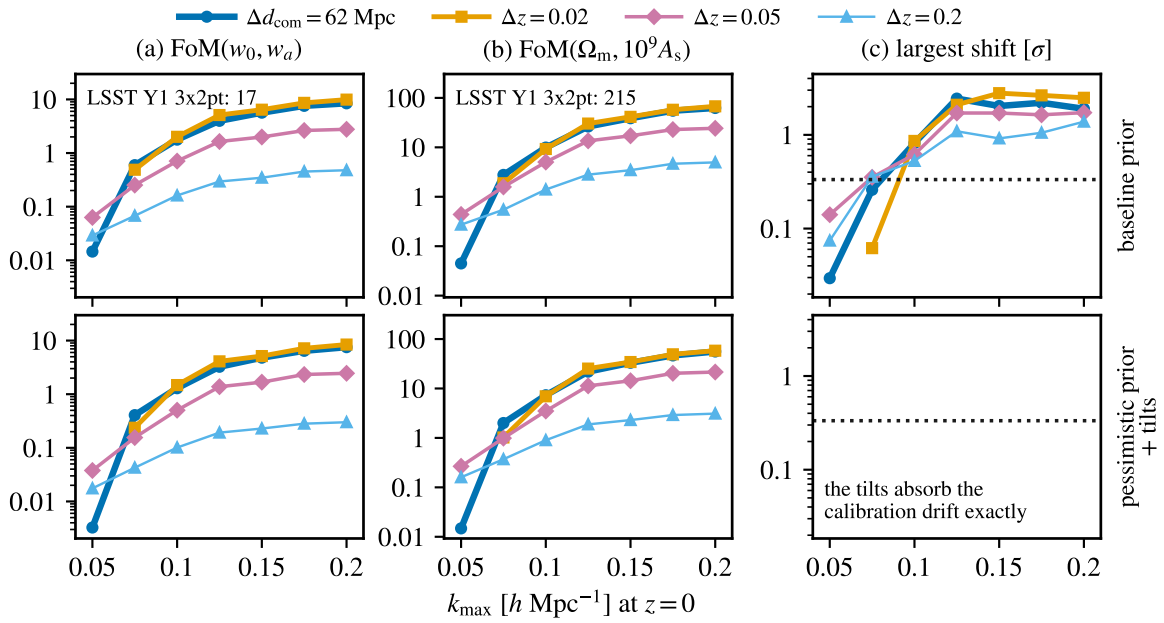


Figure 3: DESI C_ℓ^{gg} alone against $k_{\rm max}$ in two scenarios: the baseline prior with the fiducial model without tilts (top row) and the pessimistic prior with the tilts free (bottom row), named at the right. (a) FoM(w_0, w_a); (b) FoM($\Omega_m, 10^9 A_s$); (c) the largest shift of a fit over (w_0, w_a), Ω_m and A_s and over the truths outside its prior (the calibration drift), with the $\sigma/3$ line (dotted); the shifts of the truths inside the prior are in the folder's `data.json`, not drawn. Color, line width and marker: the layouts; every curve is solid with filled markers. The text in the top panels of (a) and (b) quotes the LSST Y1 3x2pt reference. The bottom row of (c) carries no curve: the tilts absorb the calibration drift exactly, as the panel states. Omitted: the free prior. On $\Delta z = 0.02$ the cut at 0.05 keeps no band. $k_{\rm max}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

SN_2, whose terms are linear, and first order for the offsets; it estimates how far an MCMC with the model would land from the truth on real data, an estimate of MCMC safety. A truth inside the fitted model's own prior—α_2, SN_2 and the offsets—can move the fit by up to about 1σ without making its error bars dishonest, so its shifts are reported, never held to a threshold; the one truth outside the prior, the calibration drift, is tested. A model holds at a cut when the tested truth moves none of (w_0, w_a), Ω_m and A_s by more than $\sigma/3$ (a shift that widens the effective error bar by only a few percent; Appendix K.1), there and at every smaller fitted cut; it breaks at the next cut. Figure 3c draws the largest tested shift. Each variant breaks at the same cut under the baseline, pessimistic and free priors.

- The baseline scenario breaks at $0.075\,h$/Mpc on the two thin layouts, where the drift moves (w_0, w_a) by 0.82 and 0.86σ at 0.1, and at 0.05 on $\Delta z = 0.05$ and 0.2 (0.35 and 0.36σ at 0.075). From 0.1 on its largest tested shift is 0.53–2.8σ.

- The pessimistic scenario passes every cut by construction: the fiducial model's tilts have the drift's shape and absorb it exactly (shifts of order $10^{-12}\sigma$), so the pass carries no

information on its own. Its accuracy rests on its honest error bars (Section 6) and on the reported in-prior shifts: up to 0.98σ at 0.075 on $\Delta d_{\rm com} = 62$ Mpc and 1.10σ at 0.1 on $\Delta z = 0.02$, both through α_2; 0.33–0.73σ from 0.125 onward on the three thinner layouts; at most 0.25σ on $\Delta z = 0.2$.

An SN_2 truth at one prior width moves no parameter by more than 0.26σ at any cut, in either scenario.

The FoM against $k_{\rm max}$. Figure 3a and b draw the FoMs of the two scenarios, with LSST Y1 3x2pt (17.26 and 214.9) for reference; the growth-tied prior, which fails the accuracy test, is in no figure, and the free prior is in the folder's `data.json`. The baseline scenario gives the larger FoMs at every cut, $\mathrm{FoM}(w_0, w_a) = 8.46$ and 9.91 on the thin layouts at $0.2\,h/\mathrm{Mpc}$, 1.12 and 1.18 times the 7.53 and 8.42 of the pessimistic scenario, with $\mathrm{FoM}(\Omega_m, 10^9 A_s)$ at most 67.5, but the calibration drift moves it by more than $\sigma/3$ from 0.075 on. The fiducial model on the pessimistic prior gives $\mathrm{FoM}(w_0, w_a) = 0.407$, 0.237, 0.155 and 0.0426 at 0.075 and 7.53, 8.42, 2.45 and 0.300 at 0.2 (layout order), and $\mathrm{FoM}(\Omega_m, 10^9 A_s) = 56.0$, 58.1, 21.5 and 3.10 at 0.2. Both scenarios sit well below LSST Y1 3x2pt in both FoMs at every cut.

A The models

This appendix first builds the prior without the tilt of Eq. (2): the kernel model, in which the exponential kernel describes the whole deviation δ_i of a bin from its group's trend. Appendix C adds the free tilt, and the kernel then describes the residual ε_i around each group's line. In the appendices b_i is the linear bias $b_{1,i}$ of bin i.

A.1 Growth-tied, free, and correlated biases

The growth-tied model sets $b_i = B_g/D(\bar{z}_i)$, with one free amplitude B_g per group. The kernel model multiplies it by a fractional deviation δ_i,

$$b_i \;=\; \frac{B_g}{D(\bar{z}_i)}\,\big(1+\delta_i\big), \tag{10}$$

$$\big\langle \delta_i\,\delta_j \big\rangle \;=\; \sigma_g^2\,\exp\!\left(-\,\frac{|\bar{z}_i-\bar{z}_j|}{L}\right), \tag{11}$$

with B_g still free (flat prior) and the prior zero across groups. In the Fisher matrix the convenient parameters are the per-bin amplitudes

$$\beta_i \;\equiv\; B_g\,\big(1+\delta_i\big), \qquad \text{fiducial } \beta_i = B_g, \tag{12}$$

so that $b_i = \beta_i/D(\bar{z}_i)$. Since $\beta_i - B_g = B_g\,\delta_i$, the prior covariance of the β_i within one group is

$$\mathsf{C} \;=\; \sigma_\beta^2\,\mathsf{R}, \qquad \sigma_\beta \;=\; \sigma_g\,B_g, \qquad \mathsf{R}_{ij} \;=\; \exp\!\left(-\,\frac{|\bar{z}_i-\bar{z}_j|}{L}\right), \tag{13}$$

with σ_β the spread of one amplitude, in units of a bias, and R the dimensionless correlation matrix.

A.2 What σ_g and L mean

The diagonal of the kernel gives the wander of a single bin, $\mathrm{Var}(\delta_i) = \sigma_g^2$. The correlation coefficient of two bins at separation $\Delta z = |\bar{z}_i - \bar{z}_j|$ is

$$r \;=\; \frac{\langle \delta_i \delta_j \rangle}{\sqrt{\langle \delta_i^2 \rangle\,\langle \delta_j^2 \rangle}} \;=\; e^{-\Delta z/L}, \tag{14}$$

set by the separation and L alone. The jump between the two bins has variance

$$\mathrm{Var}(\delta_i - \delta_j) \;=\; 2\,\sigma_g^2\left(1 - e^{-\Delta z/L}\right) \;\approx\; \frac{2\,\sigma_g^2\,\Delta z}{L} \quad (\Delta z \ll L). \tag{15}$$

At small separations the jump variance grows linearly with Δz, as for a random walk; only beyond L does the finite wander pull the deviation back. For neighbors at $\Delta z = 0.02$ with $L = 0.05$, $r = e^{-0.4} = 0.670$ and the exact rms jump is $0.81\,\sigma_g$; the small-separation form gives $0.89\,\sigma_g$, because $\Delta z/L = 0.4$ is not small.

A.3 The free group amplitude and the projection

A plain Gaussian prior with covariance C would charge for moving all β_i of a group together, but that uniform shift is the free amplitude B_g. The repair removes the uniform direction: with $\mathbf{1}$ the ones vector of the group's n bins, the prior precision of a group's amplitudes is

$$\mathsf{P} \;=\; \mathsf{C}^{-1} \;-\; \frac{\mathsf{C}^{-1}\mathbf{1}\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}}{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}}. \tag{16}$$

$\mathsf{P}\,\mathbf{1} = 0$, so a uniform shift costs nothing; a pure shape change keeps the penalty of the plain prior; P is positive semidefinite (no displacement has a negative penalty) with rank $n-1$; and P is exactly the prior obtained by sampling B_g with a flat prior and marginalizing it. Appendices F, G and H show these for two bins, by marginalization, and for n bins. Across groups the prior is zero, so the full prior precision is block diagonal.

B Which L and σ_g are physical

One case of bias evolution can be solved in closed form: a population whose members are conserved. This appendix derives it, measures its deviation from the growth-tied trend, lists what breaks conservation in a selected sample, and weighs the published measurements.

B.1 A conserved population: the bias continuity equation

Take a sample whose membership never changes, with no velocity bias, in linear theory. The matter overdensity δ_m and the galaxy-count overdensity $\delta_{\rm gal}$ then obey the same continuity equation with the same velocity field, so the excess clumping $\delta_{\rm gal} - \delta_m$ is frozen (Appendix D). With b_c the linear bias at a reference redshift z_c and $\delta_m \propto D$, the frozen difference gives [9]

$$b(z) - 1 \;=\; \big(b_c - 1\big)\,\frac{D(z_c)}{D(z)}\,: \tag{17}$$

the excess set at formation stays fixed while the matter clustering grows, so the bias decays toward one. The growth-tied shape freezes a different quantity: constant $b\,D$ means a galaxy clustering amplitude $b\,\delta_m$ that does not grow, which describes a selection that picks, at each redshift, a different and rarer set of objects.

B.2 The deviation from the growth-tied trend

Place the reference at a group's midpoint z_c, with $b_c = B/D(z_c)$. The ratio of Eq. (17) to the growth-tied bias $b_c\,D(z_c)/D(z)$ gives the deviation of Eq. (10),

$$\delta(z) \;=\; \frac{D(z)/D(z_c) - 1}{b_c}, \tag{18}$$

zero at the midpoint, positive at lower redshift and negative at higher redshift, and nearly linear across a group because D is smooth. In flat ΛCDM with $\Omega_m = 0.3$ and the project's fiducial amplitudes it is $+5.3\%$ and -4.9% at the ends of BGS ($B = 1.34$, $0.1 < z < 0.4$, $b_c = 1.5$), $+2.4\%$ and -2.2% at the ends of LRG1 ($B = 1.70$, $0.4 < z < 0.6$, $b_c = 2.2$), and $+2.6\%$ and -2.3% at the ends of LRG3 ($B = 1.70$, $0.8 < z < 1.1$, $b_c = 2.7$). Between neighbors

it changes by 0.3–0.7% at $\Delta z = 0.02$ and 0.8–1.7% at $\Delta z = 0.05$: a smooth tilt, coherent over the whole group. For comparison, the neighbor rms of Eq. (15) at $\Delta z = 0.02$ (and 0.05) is 2.4% (3.4%) for $\sigma_g = 3\%$ and $L = 0.05$, and 0.44% (0.67%) for $\sigma_g = 1\%$ and $L = 0.2$. Figure 4 draws these shapes: the conserved curve is flatter than B_g/D in every group, while the BGS flux-limit curve is much steeper, and convex.

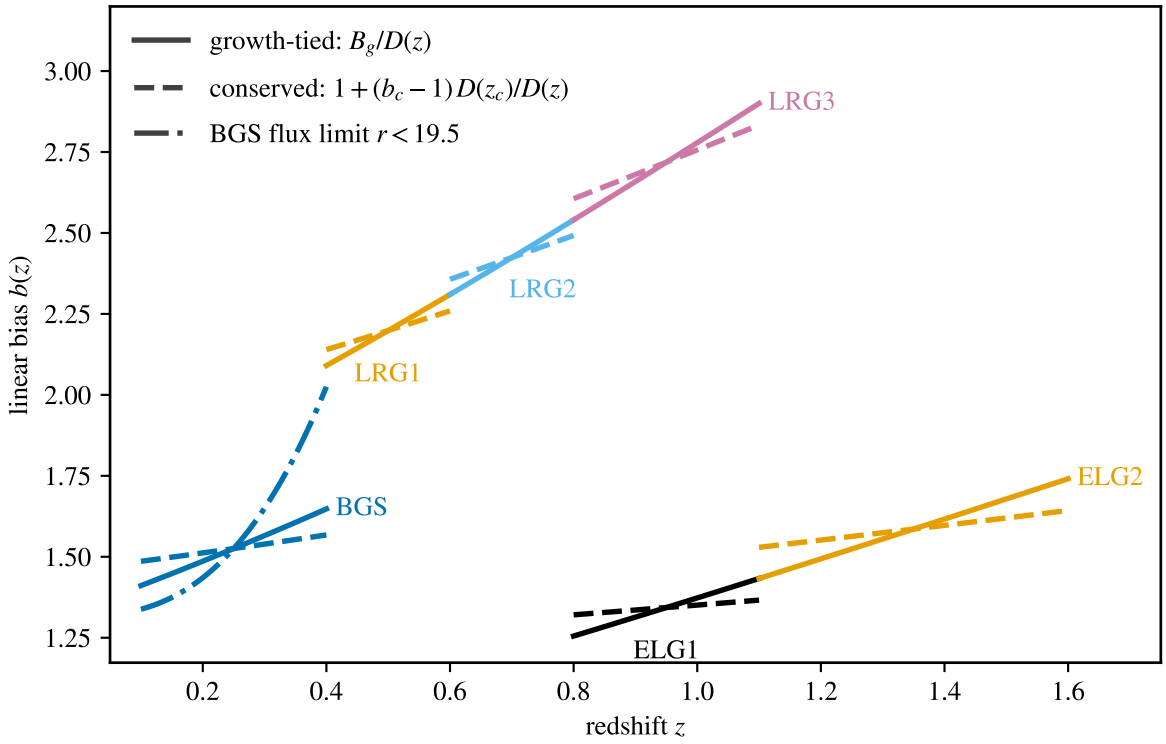


Figure 4: The bias $b(z)$ of every group. Solid: the growth-tied shape $B_g/D(z)$. Dashed: a conserved population anchored at the group midpoint (Eq. (17)). Dash-dot, on BGS only: the $r < 19.5$ flux-limit bias, anchored at the same midpoint.

B.3 What breaks conservation in a selected sample

- **Threshold drift.** A selection holds a roughly constant comoving density until the flux limit bites; beyond, it keeps only its brighter, more biased members. The change is smooth, with L comparable to the group width.

- **Color-selection features.** A spectral feature crossing a filter edge moves a color cut quickly in redshift. The sharpest is the 4000 Å break, the step in an old galaxy's rest-frame spectrum at 4000 Å, which redshift slides across the filters: a filter edge 100–200 Å wide maps to $\Delta z = \Delta\lambda/(4000\ \text{Å}) \approx 0.03$–$0.05$. The crossing is known; its amplitude in bias is unmeasured.

- **Mergers** turn two members into one.

- **Emission-line selection.** An ELG belongs to the sample while its lines are strong, so membership follows the star-formation phase.

B.4 What the measurements say

Each statement is tagged *measured* (read from the cited source), *derived* (computed here from measured inputs) or *assumed*. Equation (17) is from Fry [9]; Tegmark and Peebles [10] rederive it.

Massive red galaxies follow the conserved shape. SDSS and 2SLAQ LRGs at $0.19 < z < 0.55$, matched in color and luminosity, show no significant evolution of the clustering amplitude and merge at $7.5 \pm 2.3\%$ over that range, about 2.4% per Gyr [11] (measured). The BOSS progenitors of the SDSS LRGs merge at $2 \pm 1.5\%$ per Gyr and cluster as the continuity equation predicts [12], and CMASS samples of fixed comoving density at $0.43 < z < 0.7$ cluster like a conserved population [13] (measured). The bias of DESI-like LRGs in photometric bins of $\Delta z \approx 0.1$ stays within $\pm 2.3\%$ of $b = 1.5/D$ below $z = 0.83$, with the highest bin $6.4 \pm 6.4\%$ high [14] (measured biases, residuals derived; Figure 5).

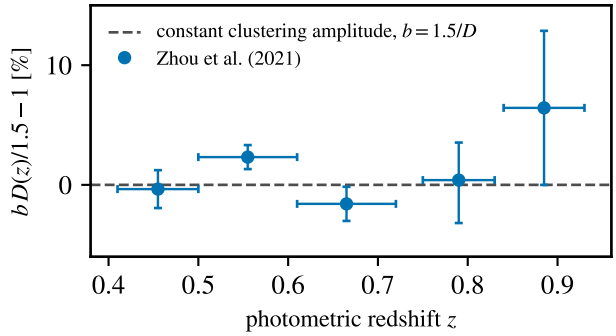


Figure 5: The measured bias of DESI-like LRGs [14] as a fractional residual around $b = 1.5/D$, with D at each photometric bin's center (flat ΛCDM, $\Omega_m = 0.3$); horizontal bars: the bin edges.

The selections drift. The DESI LRG cuts hold the comoving density near $5 \times 10^{-4}\,(h/\mathrm{Mpc})^3$ at $0.4 < z < 0.8$ and hand over to a fixed flux limit beyond [15] (measured). That limit raises the luminosity threshold by 0.27 magnitudes per $\Delta z = 0.1$ at $z \approx 1$; times the threshold-bias slope of SDSS luminosity-threshold samples, $b\,(\sigma_8/0.8) = 1.06 + 0.21\,(L_{\mathrm{th}}/L_*)^{1.12}$ [20] (0.5–1.0 in bias per magnitude near LRG luminosities; measured), it drifts the LRG3 bias by 6–11% per $\Delta z = 0.1$ on top of B/D (derived, from an r-band relation outside its calibrated range, so only the direction and order of magnitude hold). BGS Bright is magnitude limited at $r < 19.5$ [23] (measured), which drives a 10–25% bias drift across the group (derived; Appendix E).

The ELG bias shape is stable. Fits of the halo occupation distribution (HOD: how many sample galaxies live in a halo of each mass) to the DESI One-Percent ELGs in $0.8 < z < 1.1$ and $1.1 < z < 1.6$ agree within one sigma [25] (measured); at these host masses the halo bias depends weakly on mass, so the implied drift is at the percent level (derived, slope assumed).

The [OII] doublet loses signal-to-noise in windows $\Delta z \approx 0.001$–0.01 wide where it lands on sky lines, especially at $z > 1.5$ [24] (measured): the one short-separation mechanism with a measured location, with an assumed bias amplitude of 1–2%.

Nothing resolves bias below $\Delta z \approx 0.1$. The CMB-lensing tomography of the same samples [17, 18, 19], on the photometric samples of Zhou et al. [16], uses bins of about 0.15 with one free bias per bin and finds smooth models that fit. Below $\Delta z \approx 0.1$ this note rests on the continuity equation and the selection arguments.

C The memory model

C.1 Why a second model

Every deviation above except the sky-line windows is a monotonic drift: the conserved-population tilt (about 1% per $\Delta z = 0.05$), the LRG3 magnitude-limit drift (6–11% per $\Delta z = 0.1$) and the BGS flux-limit drift (10–25% across the group). The kernel model admits such a drift only at a price: to pass a 20% drift with $L \le 0.1$ it takes $\sigma_g = 10$–15%, which also allows about 10% scatter between neighbors that no identified mechanism produces. One σ_g controls both, so the kernel cannot say "large smooth drift, small local scatter". The memory model separates the two.

C.2 Why the kernel forgets

At $L = 0.05$, bins 0.15 apart share $e^{-3} \approx 0.05$ of their deviation. With $r_k = e^{-(\bar{z}_{k+1} - \bar{z}_k)/L}$ the correlation of neighbors k and $k+1$, the best guess of the next bin given all earlier ones is (Appendix H.2)

$$\mathrm{E}\big[\delta_{k+1} \mid \delta_1, \ldots, \delta_k\big] \;=\; r_k\, \delta_k : \tag{19}$$

only the nearest neighbor matters, and the influence of any one bin fades along the chain.

C.3 The model, and why it remembers

The memory model adds a tilt per group and splits the deviation into a line and a residual,

$$\delta_i \;=\; s_g\, v_i + \varepsilon_i, \qquad v_i \;=\; \bar{z}_i - \bar{z}_g, \tag{20}$$

with s_g a free tilt (flat prior, like B_g) and ε_i carrying the kernel of Eq. (11): the model of Eqs. (2) and (3). Giving the slope a Gaussian prior of variance τ^2 first (the flat prior is $\tau \to \infty$), the deviations have covariance

$$\mathrm{Cov}(\delta_i, \delta_j) \;=\; \tau^2\, v_i\, v_j \;+\; \sigma_g^2\, e^{-|\bar{z}_i - \bar{z}_j|/L}. \tag{21}$$

The new term does not decay with separation. Far beyond L the correlation coefficient tends to

$$\rho_{ij} \;\longrightarrow\; \frac{\tau^2 v_i v_j}{\sqrt{\big(\tau^2 v_i^2 + \sigma_g^2\big)\big(\tau^2 v_j^2 + \sigma_g^2\big)}} \;\longrightarrow\; \pm 1 \quad \text{when } \tau|v| \gg \sigma_g : \tag{22}$$

the two ends of a group remember each other through the shared slope. The best guess of the next bin becomes

$$\mathrm{E}\big[\delta_{k+1} \mid \delta_1, \ldots, \delta_k\big] \;=\; \hat{c} + \hat{s}_g\, v_{k+1} \;+\; r_k\,\big(\delta_k - \hat{c} - \hat{s}_g\, v_k\big), \tag{23}$$

with $(\hat{c}, \hat{s}_g)$ the level and slope of the best straight line through all past bins, weighted by C^{-1}: every past bin enters through the fitted line, and only the wiggle around it stays Markov (the universal-kriging predictor; kriging is the geostatistics name for the best linear prediction from correlated data; Appendix I). Figure 6 shows the difference on the BGS bins at $\Delta z = 0.02$: kernel draws wide enough to admit the BGS drift zigzag and forget, while memory draws are lines with small wiggles, among which the drift looks typical.

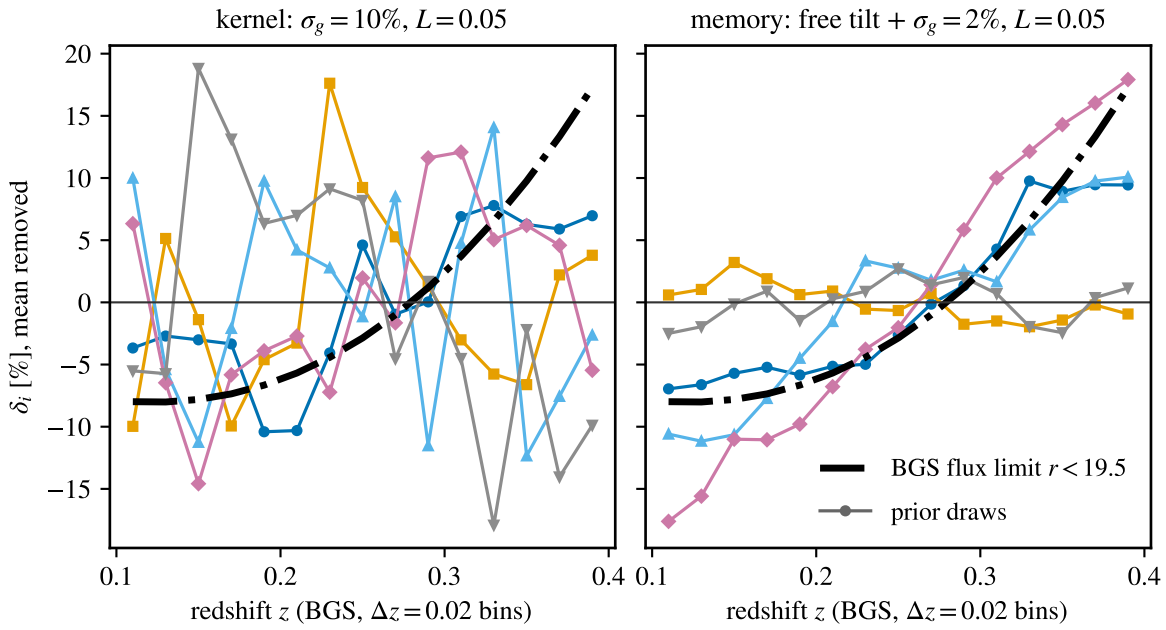


Figure 6: Random prior draws of the deviations of the 15 BGS bins at $\Delta z = 0.02$, each draw's mean removed (B_g is free). Thick dash-dot: the BGS flux-limit drift. Left: the kernel model. Right: the memory model, whose tilt is free in the model and drawn here, for display only, from a Gaussian of width 0.6 per unit redshift.

C.4 Why the memory model differs physically

A selected sample changes along a group in two ways. Coherent changes move every bin the same way: a threshold that rises with distance, the debiasing of a conserved population. Local changes move bins independently and are forgotten over a short range: the sky-line windows and filter-edge crossings of Appendix B.3. The memory model gives the first kind to the tilt and the second to the kernel. One line suffices for the coherent kind, because the drifts are smooth functions of $D(z)$ and the distance modulus. The best equal-weight line through each group's conserved deviation at the $\Delta z = 0.05$ bin centers (derived; amplitudes 1.34 for BGS, 1.70 for the LRGs and 0.84 for the ELGs) removes deviations that span 3–11% and leaves

0.02–0.16% rms, at most 0.27% (ELG2). The BGS flux-limit drift is larger and convex: it spans 22.3%, and its best line leaves 2.4% rms and 3.3% at most (Appendix E). Figure 7 shows both. In the memory model σ_g is the departure from the group's own line, curvature plus local scatter: about 0.1% for a conserved population, 1–2% for local selection features, 2–3% for the BGS curvature.

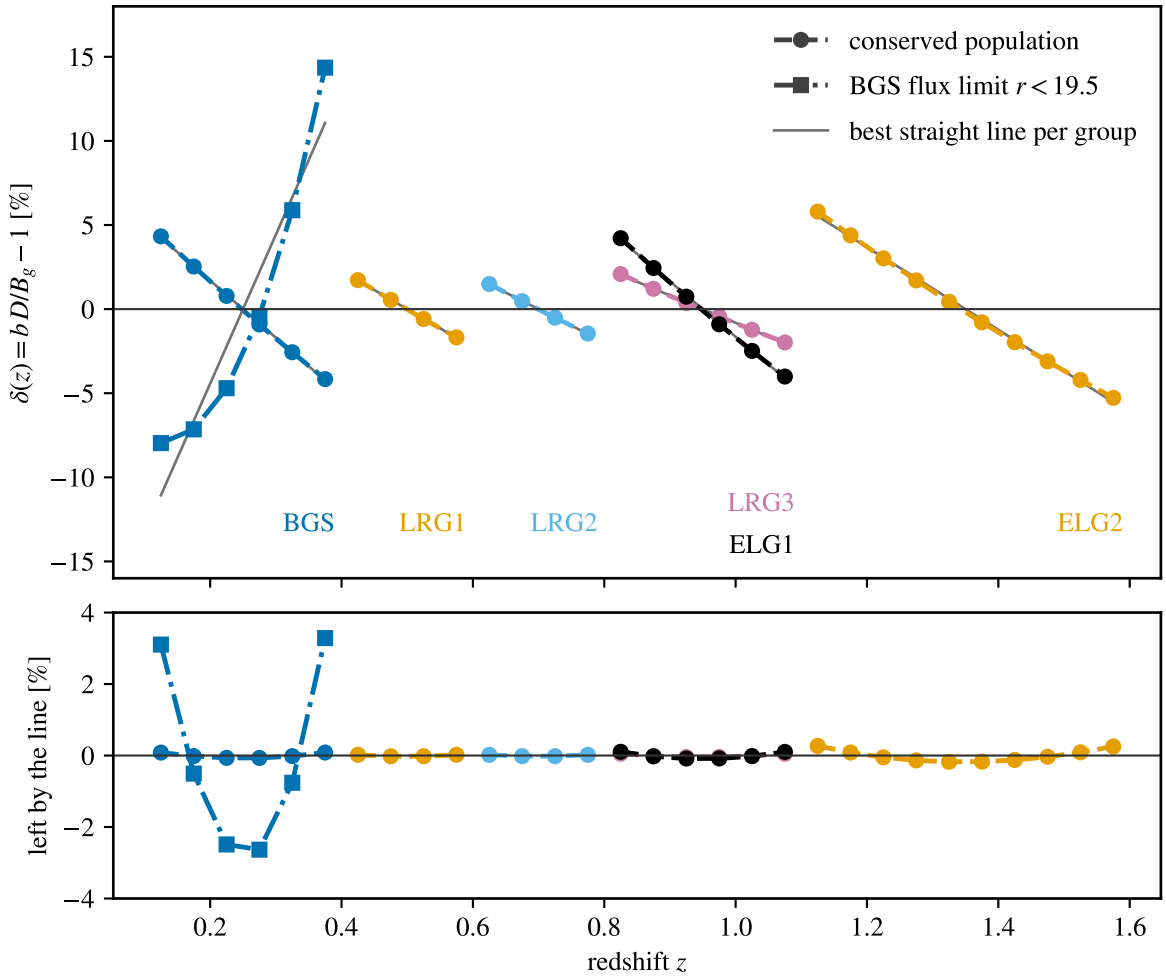


Figure 7: Top: the deviation $\delta(z) = b\,D/B_g - 1$ at the $\Delta z = 0.05$ bin centers, with B_g set so the mean deviation vanishes. Dashed with circles: the conserved population, every group. Dash-dot with squares: the BGS flux limit. Thin gray: the best straight line per curve. Bottom: what the line leaves.

C.5 What the prior charges

With flat priors on the line's level and slope, the prior precision of a group's bin amplitudes is (Appendix I)

$$\mathsf{P}_{\mathrm{mem}} \;=\; \mathsf{Q} - \mathsf{Q}\,\mathsf{U}\big(\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\big)^{-1}\mathsf{U}^{\mathsf{T}}\mathsf{Q}, \qquad \mathsf{Q} = \mathsf{C}^{-1}, \qquad \mathsf{U} = [\,\mathbf{1},\; v\,], \tag{24}$$

the construction of Eq. (16) with one more column. Its penalty is

$$\delta^{\mathsf{T}}\mathsf{P}_{\rm mem}\,\delta \;=\; \min_{c,\,s_g}\big(\delta - c\,\mathbf{1} - s_g\,v\big)^{\mathsf{T}}\mathsf{Q}\big(\delta - c\,\mathbf{1} - s_g\,v\big)\,: \tag{25}$$

the prior subtracts the best straight line and charges only the residual. A monotonic drift of any size costs nothing; bumps around the line still cost, and still forget over L. Q is tridiagonal (Appendix H), while the subtracted term has rank two and is dense, because the fitted line listens to every bin. $\mathsf{P}_{\rm mem}\mathsf{U} = 0$, $\mathsf{P}_{\rm mem}$ is positive semidefinite with rank $n-2$, and a group of one or two bins has $\mathsf{P}_{\rm mem} = 0$.

D The bias continuity equation, step by step

The matter overdensity is $\delta_m = (\rho_m - \bar\rho_m)/\bar\rho_m$, with $\bar\rho_m(t)$ the mean density; the galaxy-count overdensity $\delta_{\rm gal}$ is defined the same way. Conservation of mass and of galaxy number gives, in comoving coordinates and to linear order (the convective term $\mathbf{v}\cdot\nabla\delta$ is second order),

$$\frac{\partial\delta_m}{\partial t} \;=\; -\frac{1}{a}\,\nabla\cdot\mathbf{v}_m, \qquad \frac{\partial\delta_{\rm gal}}{\partial t} \;=\; -\frac{1}{a}\,\nabla\cdot\mathbf{v}_{\rm gal}, \tag{26}$$

with t cosmic time and $\mathbf{v}_m$, $\mathbf{v}_{\rm gal}$ the peculiar velocities. On large scales galaxies fall in the same potential as the matter and carry no pressure of their own, so $\mathbf{v}_{\rm gal} = \mathbf{v}_m$ (no velocity bias; galaxies as test particles in Tegmark and Peebles [10]). Subtracting the two equations and integrating from z_c,

$$\delta_{\rm gal}(z) - \delta_m(z) \;=\; \delta_{\rm gal}(z_c) - \delta_m(z_c)\,: \tag{27}$$

the excess clumping keeps its initial value. With $\delta_{\rm gal}(z_c) = b_c\,\delta_m(z_c)$ and $\delta_m(z) = \delta_m(z_c)\,D(z)/D(z_c)$,

$$\delta_{\rm gal}(z) \;=\; \delta_m(z)\left[\,1 + \big(b_c - 1\big)\,\frac{D(z_c)}{D(z)}\,\right], \tag{28}$$

and dividing by $\delta_m(z)$ gives Eq. (17); dividing by the growth-tied bias gives Eq. (18). As D grows the frozen term is divided by an ever larger number, so $b \to 1$: gravity debiases a conserved sample.

Tegmark and Peebles also allow stochastic bias, with b the ratio of the galaxy to the matter fluctuation amplitude and ρ_{gm} the correlation coefficient of the two fields. The same argument evolves both: with D normalized to one today and the subscript zero marking today,

$$b_0 = \big[(1-D)^2 + 2D(1-D)\,b\,\rho_{gm} + D^2 b^2\big]^{1/2}, \qquad \rho_{gm,0} = \frac{(1-D) + D\,b\,\rho_{gm}}{b_0}, \tag{29}$$

with b and ρ_{gm} the values at the epoch of growth factor D. Both approach one as clustering grows, and for $\rho_{gm} = 1$ Eq. (29) collapses to $b_0 - 1 = (b-1)\,D$, Eq. (17) again. For a DESI sample the assumption that breaks is conservation (Appendix B.3); no velocity bias and linear theory hold well on the scales kept. The script of Appendix L integrates Eq. (26) and recovers Eq. (17) to 2×10^{-9}.

E The BGS flux-limit bias model, step by step

This appendix derives the BGS drift of the baseline prior: a 22.3% span, a convex shape, and a 2.4% rms residual around the best line (the function `bgs_flux_limit_bias` of `plots/make_theory_data.py`).

E.1 The four steps

Step 1: the flux limit becomes a luminosity threshold. BGS Bright keeps every galaxy brighter than $r = 19.5$ [23] (r the magnitude band). The apparent magnitude r and the absolute magnitude M are related by $r = M + 25 + 5\log_{10}[d_L/(h^{-1}\,\mathrm{Mpc})] + K(z)$, with d_L the luminosity distance and $K(z)$ the K-correction, the magnitude correction for the part of the redshifted spectrum that falls outside the observed band [22]. With $K = 0$ the faintest absolute magnitude that passes at redshift z is

$$M_{\rm lim}(z) - 5\log_{10} h \;=\; 19.5 - 25 - 5\log_{10}\frac{d_L(z)}{h^{-1}\,\mathrm{Mpc}}, \qquad d_L(z) \;=\; (1+z)\,\chi(z), \tag{30}$$

with $\chi(z)$ the comoving distance of flat ΛCDM with $\Omega_m = 0.3$ and h the Hubble constant in units of 100 km/s/Mpc.

Steps 2 and 3: each bin is a threshold sample, and the threshold sets the bias. At redshift z the sample holds every galaxy with $M < M_{\rm lim}(z)$, a threshold that brightens with distance. The SDSS luminosity-threshold samples at $z < 0.25$ [20] give

$$b(>L_{\rm th}) \;=\; 1.06 + 0.21\,(L_{\rm th}/L_*)^{1.12}, \qquad L_{\rm th}/L_* \;=\; 10^{-0.4\,(M_{\rm lim}-M_*)}, \tag{31}$$

with $M_* - 5\log_{10} h = -20.44$ the characteristic magnitude of the SDSS r-band luminosity function [21]. The model applies the fit at every redshift, without evolution, and reads the published $b\,(\sigma_8/0.8)$ as b.

Step 4: the deviation from the growth-tied shape. Because B_g is free, only the shape of $b(z)$ matters; the amplitude makes the mean deviation over the group's bin centers vanish:

$$\delta(\bar z_i) \;=\; \frac{b(\bar z_i)\,D(\bar z_i)}{B} - 1, \qquad B \;=\; \frac{1}{n}\sum_{j=1}^{n} b(\bar z_j)\,D(\bar z_j). \tag{32}$$

At the six $\Delta z = 0.05$ bin centers of BGS, $z = 0.125$ to 0.375, the threshold brightens from -18.56 to -21.24, so $L_{\rm th}/L_*$ grows from 0.18 to 2.10 and the bias from 1.09 to 1.54, while D falls from 0.937 to 0.824. The deviation runs from -8.0% to $+14.4\%$, the 22.3% span of Figure 7.

E.2 Why the drift is convex

The bias rises by 41% while D falls by 12%, so the deviation inherits the shape of the bias curve, which is flat at low redshift and steep at high redshift: below L_* the term $0.21\,(L_{\rm th}/L_*)^{1.12}$ is small against 1.06 (0.03 at the first center), the threshold crosses L_* near $z = 0.27$, and beyond it the term rises exponentially (0.48 at the last center). A straight line through a

convex curve leaves a U-shaped residual, $(+3.1, -0.5, -2.5, -2.6, -0.8, +3.3)\%$ here. Adding a quadratic term cuts the residual from 2.4% to 0.1% rms, so the 2.4% is the curvature of the drift, the one part a free tilt cannot absorb.

E.3 What the model leaves out

None of the omissions changes the qualitative shape, a smooth, monotonic, convex rise toward the far edge of the group.

- **Evolution of the threshold-bias relation.** The bias at fixed luminosity grows with redshift (Eq. (17)), which steepens the drift; M_* brightens by 1–2 magnitudes per unit redshift [21], which flattens it, but only by 0.3–0.6 magnitudes at the last center against the 2.68 of the threshold.
- **No K-correction.** A positive K, 0.37 magnitudes at $z = 0.4$ for a spectrum flat in wavelength and more for a red one [22], makes the true threshold brighter at high redshift and steepens the drift.
- **Only $r < 19.5$.** BGS Faint, at $19.5 < r < 20.175$ [23], lowers the threshold by 0.675 magnitudes and flattens the drift; the fiber-magnitude cut and the color selection change the galaxy mix in a way Eq. (31) cannot follow.

The model is an order-of-magnitude estimate, not a prediction of the measured DESI BGS bias. Its reliable content is the shape of the drift and its size to a factor of order unity, which is what the baseline prior uses: $\sigma_g = 3\%$ for BGS, from the 2–3% curvature.

F The projected precision for two bins

Everything essential already happens with two bins, of amplitudes β_1 and β_2, common fiducial B, spread σ_β and correlation r:

$$\mathsf{C} \;=\; \sigma_\beta^2 \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}, \qquad \mathsf{C}^{-1} \;=\; \frac{1}{\sigma_\beta^2\,(1-r^2)} \begin{pmatrix} 1 & -r \\ -r & 1 \end{pmatrix}. \tag{33}$$

The penalty of a displacement $\Delta\beta = (\beta_1 - B,\ \beta_2 - B)^{\mathsf{T}}$ is $\Delta\beta^{\mathsf{T}}\,\mathsf{C}^{-1}\Delta\beta$. A uniform shift, $\Delta\beta = c\,(1,1)^{\mathsf{T}}$, costs

$$\frac{2c^2}{\sigma_\beta^2\,(1+r)} \;\neq\; 0, \tag{34}$$

but it is a change of B_g, which the growth-tied model leaves free. A shape change, $\Delta\beta = d\,(1,-1)^{\mathsf{T}}$, costs

$$\frac{2d^2}{\sigma_\beta^2\,(1-r)}, \tag{35}$$

which grows without bound as $r \to 1$. At $r = 0.670$ (Appendix A.2) the shape costs $2/(1-r) = 2/0.330 = 6.1$ in units of d^2/σ_β^2 and the uniform shift $2/(1+r) = 2/1.670 = 1.2$ in units of c^2/σ_β^2. Both rows of C^{-1} have the same sum, so $\mathsf{C}^{-1}\mathbf{1} = [\sigma_\beta^2(1+r)]^{-1}(1,1)^{\mathsf{T}}$ and Eq. (16) gives

$$\mathsf{P} \;=\; \frac{1}{2\sigma_\beta^2(1-r)} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad \Delta\beta^{\mathsf{T}}\mathsf{P}\,\Delta\beta \;=\; \frac{(\beta_1-\beta_2)^2}{2\sigma_\beta^2(1-r)}. \tag{36}$$

The denominator is the prior variance of the jump, $2\sigma_\beta^2(1-r)$: *the prior constrains the jump, not the level.*

G The projection as a marginalization

Equation (16) removes the prior's information about one direction, the uniform shift: the projected penalty of $\Delta\beta$ is the plain penalty of what remains after subtracting the best uniform shift, the generalized-least-squares mean $\hat{c} = \mathbf{1}^\mathsf{T}\mathsf{C}^{-1}\Delta\beta/\mathbf{1}^\mathsf{T}\mathsf{C}^{-1}\mathbf{1}$: the average that weights correlated bins by their inverse covariance. The derivation below shows that this is the correct prior, not just one possible repair.

G.1 Marginalizing a flat-prior amplitude

Model the n amplitudes of a group as

$$\beta \;=\; B\,\mathbf{1} + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \mathsf{C}), \tag{37}$$

with a flat prior on the scalar B: the model of Eq. (10), linearized around the fiducial. Integrating B out,

$$p(\beta) \;\propto\; \int dB\, \exp\Big[-\tfrac{1}{2}\big(\beta - B\,\mathbf{1}\big)^\mathsf{T}\mathsf{C}^{-1}\big(\beta - B\,\mathbf{1}\big)\Big]. \tag{38}$$

With $S = \mathbf{1}^\mathsf{T}\mathsf{C}^{-1}\mathbf{1}$, completing the square gives

$$\big(\beta - B\mathbf{1}\big)^\mathsf{T}\mathsf{C}^{-1}\big(\beta - B\mathbf{1}\big) \;=\; S\,\big(B - B_*\big)^2 \;+\; \beta^\mathsf{T}\mathsf{C}^{-1}\beta \;-\; \frac{\big(\mathbf{1}^\mathsf{T}\mathsf{C}^{-1}\beta\big)^2}{S}, \qquad B_* = \frac{\mathbf{1}^\mathsf{T}\mathsf{C}^{-1}\beta}{S}. \tag{39}$$

The B integral contributes only the constant $\sqrt{2\pi/S}$, and what survives is

$$p(\beta) \;\propto\; \exp\Big[-\tfrac{1}{2}\,\beta^\mathsf{T}\mathsf{P}\,\beta\Big], \qquad \mathsf{P} \;=\; \mathsf{C}^{-1} - \frac{\mathsf{C}^{-1}\mathbf{1}\,\mathbf{1}^\mathsf{T}\mathsf{C}^{-1}}{\mathbf{1}^\mathsf{T}\mathsf{C}^{-1}\mathbf{1}}, \tag{40}$$

exactly Eq. (16). In matrix language, the exponent of Eq. (38) is a Gaussian in (β, B) with precision

$$\mathsf{F}_{(\beta,B)} \;=\; \begin{pmatrix} \mathsf{C}^{-1} & -\,\mathsf{C}^{-1}\mathbf{1} \\ -\,\mathbf{1}^\mathsf{T}\mathsf{C}^{-1} & \mathbf{1}^\mathsf{T}\mathsf{C}^{-1}\mathbf{1} \end{pmatrix}, \tag{41}$$

and removing B leaves the Schur complement $\mathsf{P} = \mathsf{F}_{\beta\beta} - \mathsf{F}_{\beta B}\,\mathsf{F}_{BB}^{-1}\,\mathsf{F}_{B\beta}$: the block corrected downward by what B could absorb. P is the statement "*B_g is sampled with a flat prior and marginalized*", folded into one matrix.

H Many bins: the tridiagonal inverse

H.1 Three bins: total correlation versus direct links

With the covariance $\mathsf{C}_{ij} = \sigma_\beta^2 \exp(-|\bar{z}_i - \bar{z}_j|/L)$ of Eq. (13), positive definite for distinct redshifts, three evenly spaced bins with neighbor correlation r give

$$\mathsf{C} \;=\; \sigma_\beta^2 \begin{pmatrix} 1 & r & r^2 \\ r & 1 & r \\ r^2 & r & 1 \end{pmatrix}, \qquad \mathsf{C}^{-1} \;=\; \frac{1}{\sigma_\beta^2\,(1-r^2)} \begin{pmatrix} 1 & -r & 0 \\ -r & 1+r^2 & -r \\ 0 & -r & 1 \end{pmatrix}. \tag{42}$$

Bins 1 and 3 *are* correlated (r^2 in C), yet their entry in C^{-1} is zero: C gives the total correlation, what δ_1 says about δ_3 when nothing else is known, while C^{-1} gives the direct one, what δ_1 still says once δ_2 is known, which is nothing. The exponential kernel makes this exact because its correlations multiply along the chain,

$$e^{-(\bar{z}_3-\bar{z}_1)/L} \;=\; e^{-(\bar{z}_3-\bar{z}_2)/L}\, e^{-(\bar{z}_2-\bar{z}_1)/L}; \tag{43}$$

a Gaussian kernel $e^{-\Delta z^2/L^2}$ does not factorize, and its inverse is dense.

H.2 The chain of links

Because correlations multiply along the chain, the joint density of the deviations factorizes into neighbor conditionals,

$$\begin{aligned} p(\delta_1,\dots,\delta_n) \;&=\; p(\delta_1)\, p(\delta_2 \mid \delta_1)\cdots p(\delta_n \mid \delta_{n-1}), \\ \big\langle \delta_{k+1} \mid \delta_k \big\rangle = r_k\, \delta_k, \qquad &\mathrm{Var}\big(\delta_{k+1} \mid \delta_k\big) = \sigma_g^2\,\big(1-r_k^2\big), \end{aligned} \tag{44}$$

with $r_k = e^{-(\bar{z}_{k+1}-\bar{z}_k)/L}$. Splitting $\delta_{k+1} = \lambda\,\delta_k + e$ with e uncorrelated with δ_k gives $\lambda = r_k$ and $\mathrm{Var}(e) = \sigma_g^2(1-r_k^2)$, and earlier bins add nothing: for $j \le k$,

$$\mathrm{Cov}(e,\delta_j) \;=\; \sigma_g^2\, e^{-(\bar{z}_k-\bar{z}_j)/L}\left[e^{-(\bar{z}_{k+1}-\bar{z}_k)/L} - r_k\right] \;=\; 0. \tag{45}$$

Taking $-2\ln p$ gives the penalty as a sum over neighbor links,

$$\delta^{\mathsf{T}}\big(\sigma_g^2\mathsf{R}\big)^{-1}\delta \;=\; \frac{\delta_1^2}{\sigma_g^2} \;+\; \sum_{k=1}^{n-1} \frac{\big(\delta_{k+1} - r_k\,\delta_k\big)^2}{\sigma_g^2\,\big(1-r_k^2\big)}\,: \tag{46}$$

directly only neighbors talk; in total the influence propagates down the chain (the Markov property).

H.3 The tridiagonal inverse

Collecting the coefficient of every product $\delta_i\delta_j$ in Eq. (46) reads off the precision matrix. For the amplitudes ($\Delta\beta = B_g\,\delta$, $\sigma_\beta = \sigma_g B_g$) it is tridiagonal, nonzero only on the diagonal and on the two diagonals next to it,

$$\sigma_\beta^2\,\big(\mathsf{C}^{-1}\big)_{kk} = \begin{cases} \dfrac{1}{1-r_1^2}, & k=1, \\[2mm] \dfrac{1}{1-r_{k-1}^2} + \dfrac{r_k^2}{1-r_k^2}, & 1<k<n, \\[2mm] \dfrac{1}{1-r_{n-1}^2}, & k=n, \end{cases} \qquad \sigma_\beta^2\,\big(\mathsf{C}^{-1}\big)_{k,k+1} = \frac{-\,r_k}{1-r_k^2}\,: \tag{47}$$

the off-diagonal entry is the link of pair k, and the diagonal entry of a bin collects the links touching it. For even spacing the interior diagonal becomes $(1+r^2)/(1-r^2)$ and

$$\mathsf{C}^{-1} \;=\; \frac{1}{\sigma_\beta^2\,(1-r^2)} \begin{pmatrix} 1 & -r & & & \\ -r & 1+r^2 & -r & & \\ & -r & \ddots & \ddots & \\ & & \ddots & 1+r^2 & -r \\ & & & -r & 1 \end{pmatrix}, \tag{48}$$

which for $n = 3$ is Eq. (42).

H.4 The projection, group by group, and its limits

Each group receives the projection of Eq. (16) built from its own C, and the total prior precision is block diagonal. The kernel model contains both extremes. As $L \to \infty$, every $r_k \to 1$, the jump variance of every neighbor pair vanishes, and only the uniform shift is left: the growth-tied model, whatever σ_g. As $\sigma_g \to \infty$ at fixed L, $\mathsf{P} \to 0$: the free model.

I The memory projection, step by step

A group has n bins, C is the kernel covariance of the residuals, $\mathsf{Q} = \mathsf{C}^{-1}$, $\mathsf{U} = [\,\mathbf{1},\ v\,]$ the $n \times 2$ design matrix of Eq. (24) with v the centered redshifts, and $\theta = (c, s_g)^{\mathsf{T}}$ a line's level and slope. (For the bin amplitudes every covariance carries a factor B_g^2 that cancels in each identity below.)

I.1 The line fit, the split, and the marginalization

Minimizing $(\delta - \mathsf{U}\theta)^{\mathsf{T}}\mathsf{Q}\,(\delta - \mathsf{U}\theta)$ over the line gives the normal equations

$$\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\,\hat{\theta} \;=\; \mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\delta, \qquad \hat{\theta} = (\hat{c}, \hat{s}_g)^{\mathsf{T}}, \tag{49}$$

with $\mathsf{U}^{\mathsf{T}}\mathsf{Q}\mathsf{U}$ positive definite whenever the group has two distinct redshifts. Writing $\delta = \mathsf{U}\hat{\theta} + \delta_\perp$, the residual obeys $\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\delta_\perp = 0$, the cross terms vanish, and the penalty splits,

$$\delta^{\mathsf{T}}\mathsf{Q}\,\delta \;=\; \hat{\theta}^{\mathsf{T}}\big(\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\big)\hat{\theta} \;+\; \delta_\perp^{\mathsf{T}}\mathsf{Q}\,\delta_\perp, \qquad \delta^{\mathsf{T}}\mathsf{P}_{\mathrm{mem}}\,\delta \;=\; \delta_\perp^{\mathsf{T}}\mathsf{Q}\,\delta_\perp : \tag{50}$$

the projection keeps the residual's penalty and discards the line's. A pure line has $\delta_\perp = 0$, so $\mathsf{P}_{\mathrm{mem}}\mathsf{U} = 0$; the residuals obey two linear constraints, so the rank is $n - 2$; and $\delta_\perp^{\mathsf{T}}\mathsf{Q}\,\delta_\perp \geq 0$ makes it positive semidefinite.

Marginalization gives the same matrix. With $\delta \sim \mathcal{N}(\mathsf{U}\theta, \mathsf{C})$ and $\theta \sim \mathcal{N}(0, \Lambda)$, $\Lambda = \mathrm{diag}(W^2, \tau^2)$ the prior variances of the level and the slope, the marginal covariance $\mathsf{C} + \mathsf{U}\Lambda\mathsf{U}^{\mathsf{T}}$ inverts by the Woodbury identity,

$$\big(\mathsf{C} + \mathsf{U}\Lambda\mathsf{U}^{\mathsf{T}}\big)^{-1} \;=\; \mathsf{Q} - \mathsf{Q}\,\mathsf{U}\big(\Lambda^{-1} + \mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\big)^{-1}\mathsf{U}^{\mathsf{T}}\mathsf{Q}, \tag{51}$$

which becomes $\mathsf{P}_{\mathrm{mem}}$ as $\Lambda^{-1} \to 0$, with a leading correction falling as $1/W^2$.

I.2 The null-space form

Let the columns of the $n \times (n-2)$ matrix N span the combinations of bins that no straight line can produce, $\mathsf{N}^{\mathsf{T}}\mathsf{U} = 0$. Then

$$\mathsf{Q} - \mathsf{Q}\,\mathsf{U}\big(\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\big)^{-1}\mathsf{U}^{\mathsf{T}}\mathsf{Q} \;=\; \mathsf{N}\big(\mathsf{N}^{\mathsf{T}}\mathsf{C}\,\mathsf{N}\big)^{-1}\mathsf{N}^{\mathsf{T}}. \tag{52}$$

Both sides annihilate U and map CN to N, and the columns of CN and U together span the whole space (if $\mathsf{CN}x + \mathsf{U}y = 0$, multiplying by N^{T} gives $x = 0$, then $y = 0$), so the two matrices are equal. The prior reads off the combinations $\mathsf{N}^{\mathsf{T}}\delta$ that no line can make and charges each by the inverse of its own prior covariance. With the single column $\mathsf{U} = \mathbf{1}$ the same identity gives Eq. (16).

I.3 Worked examples

For $n = 2$, U is square and invertible and the projection vanishes; for $n = 1$, N has no columns. For three evenly spaced bins with spacing Δ and $r = e^{-\Delta/L}$, $v = (-\Delta, 0, \Delta)^{\mathsf{T}}$ and the only direction orthogonal to $\mathbf{1}$ and v is the discrete curvature $\mathsf{N} = (1, -2, 1)^{\mathsf{T}}$, with prior variance $\mathsf{N}^{\mathsf{T}}\mathsf{C}\,\mathsf{N} = 2\sigma_g^2(1-r)(3-r)$, so

$$\delta^{\mathsf{T}}\mathsf{P}_{\mathrm{mem}}\,\delta \;=\; \frac{\big(\delta_1 - 2\delta_2 + \delta_3\big)^2}{2\,\sigma_g^2\,(1-r)(3-r)}\,: \tag{53}$$

the memory model charges only the curvature, where the kernel model of Eq. (42) taxes a drift as heavily as a zigzag of the same step.

I.4 The universal-kriging predictor

Predict bin $k+1$ from the first $k \geq 2$ bins, with flat priors on θ. With C_k the past kernel block, U_k the past rows of U, $u_{k+1} = (1, v_{k+1})^{\mathsf{T}}$ and γ the kernel covariances between bin $k+1$ and the past,

$$\mathrm{E}\big[\delta_{k+1} \mid \delta_{1..k}, \theta\big] \;=\; u_{k+1}^{\mathsf{T}}\theta \;+\; \gamma^{\mathsf{T}}\mathsf{C}_k^{-1}\big(\delta_{1..k} - \mathsf{U}_k\,\theta\big). \tag{54}$$

With a flat prior, θ is replaced by the fit $\hat{\theta}$ of Eq. (49) on the past block, and by Eq. (43) $\gamma^{\mathsf{T}}\mathsf{C}_k^{-1} = r_k\,e_k^{\mathsf{T}}$, with e_k the k-th unit vector. Substituting gives Eq. (23). With a Gaussian slope of variance τ^2, dropping the kernel term far beyond L gives the correlation of Eq. (22).

J How the prior enters the Fisher matrix

A Gaussian prior adds its precision to the Fisher matrix,

$$F \;=\; F_{\mathrm{data}} \;+\; F_{\mathrm{priors}}, \tag{55}$$

with one block per prior. $\mathsf{P}_{\mathrm{mem}}$ of each group is added to the rows and columns of that group's amplitudes β_i for the baseline and pessimistic priors; the growth-tied prior instead sums each group's amplitude columns into one (Appendix M.4). A bin that lost all its band powers to the scale cut has zero rows in F_{data}: under the free prior its amplitude is dropped, while a correlated prior ties it to its neighbors and keeps the matrix invertible as long as the group's data fix its level and tilt.

K The accuracy test

This appendix gives the method of Section 6 and of the shift estimates of Section 7.

K.1 The shift formula

Let m be the expected data vector (the band powers), C_d its covariance, D the matrix of derivatives of m with respect to the parameters (sans-serif, not the growth factor D), and $F = \mathsf{D}^{\mathsf{T}}\mathsf{C}_d^{-1}\mathsf{D} + F_{\mathrm{priors}}$ the Fisher matrix of one fit, its priors included (Appendix J). If the truth moves the data vector by δm, the maximum of the posterior moves, to first order, by

$$\delta p \;=\; F^{-1}\mathsf{D}^{\mathsf{T}}\mathsf{C}_d^{-1}\,\delta m\,: \tag{56}$$

the score (the gradient of the log posterior) at the old maximum becomes $\mathsf{D}^\mathsf{T}\mathsf{C}_d^{-1}\delta m$ instead of zero, and the curvature F converts it into a displacement. With C_w the fit's marginalized 2×2 covariance of (w_0, w_a) and δw the (w_0, w_a) entries of δp, the shift

$$\Delta_w \;=\; \sqrt{\delta w^\mathsf{T}\, \mathsf{C}_w^{-1}\, \delta w} \tag{57}$$

is the displacement of the dark-energy ellipse in units of its own size; Ω_m and A_s shifts are $|\delta p|/\sigma_p$. The threshold is 1/3: a shift of a third of an error bar, added in quadrature to the noise, widens the effective error to $(1 + 1/9)^{1/2} = 1.05$ of its value, a few percent. Ellipse and response use the same F, so a rigid model is punished twice: it absorbs little of the drift, and its ellipse is small.

A drift of the bin biases enters as $\delta m = \sum_i (\partial m/\partial \beta_i)\, B_g\, \delta_i$, with β_i the amplitudes of Eq. (12); under the one-loop model the column is that of Eq. (66) divided by $D(\bar{z}_i)$, so a drift of b_1 drags b_2 and b_{s^2} along. The growth-tied prior is evaluated in its own parameter basis. For the free prior the bias derivatives span the drift exactly, so the cosmology does not move, and that accuracy is bought with the information the prior gives up.

K.2 The drift laws

The conserved-population drift applies to every group: the deviation of Eq. (18), anchored at the group's mean redshift, nearly a straight line of a few percent across each group. The BGS flux-limit drift applies to BGS alone: the convex deviation of Appendix E, a 22% span whose best line leaves a 2.4% rms curvature. Both come from `plots/make_theory_data.py`; only their shapes matter, because the free amplitudes remove the level.

K.3 Random truths

A random truth sets the bias of bin i in group g to

$$b_i \;=\; \frac{B_g}{D(\bar{z}_i)}\left(1 + \delta_i^{\mathrm{drift}} + \varepsilon_i\right) \tag{58}$$

at the fiducial cosmology, with $\delta_i^{\mathrm{drift}}$ the known drift (the flux limit in BGS, the conserved population elsewhere) and ε_i one draw from $\mathcal{N}(0, \Sigma)$, whose covariance $\Sigma_{ij} = \sigma_g^2 e^{-|\bar{z}_i - \bar{z}_j|/L}$ inside a group, zero across groups, is the kernel of the pessimistic prior. Every truth is fitted under the growth-tied and the pessimistic prior, each centered on the growth-tied shape. No fit is told which truth it faces.

Equation (56) is linear in δm, and δm in the bias deviations, so the (w_0, w_a) rows of the response collapse into one $2 \times n_{\mathrm{bins}}$ matrix A per prior, layout and variant,

$$\delta w \;=\; \mathsf{A}\,\delta, \qquad \delta_i \;=\; \delta_i^{\mathrm{drift}} + \varepsilon_i, \tag{59}$$

and the average over the whole family of truths is exact (Appendix K.4). The band powers are quadratic in the bin biases, so the dropped second-order term is a few percent of the shift for 1–3% residuals and perhaps 10–20% for the 22% BGS drift. Where the fit's prior equals the truth's scatter, the average shift is not a bias: that fit's ellipse is wider than the growth-tied one precisely because it marginalizes over the scatter the truth has.

K.4 The test statistic

A fit is correctly specified when its ellipse covers the truth with the stated probability. The squared distance of the fitted (w_0, w_a), $\hat{w}$, from the truth $w_{\rm true}$ in units of the ellipse is $\chi^2_w = (\hat{w} - w_{\rm true})^{\mathsf{T}} \mathsf{C}_w^{-1} (\hat{w} - w_{\rm true})$. Averaged over the bias history and the data noise,

$$\langle \chi^2_w \rangle \;=\; \Delta^2_{\rm drift} \;+\; \mathrm{tr}\Big(\mathsf{C}_w^{-1} \mathsf{A}\, \Sigma\, \mathsf{A}^{\mathsf{T}}\Big) \;+\; \mathrm{tr}\big(\mathsf{C}_w^{-1} \mathsf{C}_w^{\rm noise}\big)\,: \tag{60}$$

the shift $\Delta_{\rm drift}$ of the known drift alone, squared; the scatter that the fit's bias parameters, held back by their prior, did not follow; and the noise. $\mathsf{C}_w^{\rm noise}$ is the share of C_w from the data noise and the Gaussian priors other than the bias prior; the cross terms average to zero. Eq. (60) is evaluated in closed form, the exact average over infinitely many universes; no finite set of draws enters. The quoted value is $\eta = \sqrt{\langle \chi^2_w \rangle / 2}$. An honest ellipse gives $\langle \chi^2_w \rangle = \mathrm{tr}(\mathsf{C}_w^{-1} \mathsf{C}_w) = 2$, one unit per parameter, and growing both axes by η restores that.

Coverage of one parameter. For one parameter p, in units of its marginalized error σ_p,

$$\langle \chi^2_p \rangle \;=\; \mu^2 \;+\; \mathsf{A}_p^{\mathsf{T}} \Sigma\, \mathsf{A}_p \;+\; \nu, \tag{61}$$

with μ the shift of the known drift alone, A_p the row of the response that carries the bias deviations into p, divided by σ_p, and ν the noise term, the (p,p) entry of $F^{-1} F_0 F^{-1}$ over σ_p^2, where $F_0 = \mathsf{D}^{\mathsf{T}} \mathsf{C}_d^{-1} \mathsf{D} + F_{\rm other}$ holds the data and the Gaussian priors other than the bias prior. The bias prior adds a positive semidefinite precision, $F \succeq F_0$, so $F^{-1} F_0 F^{-1} \preceq F^{-1}$ and $\nu \le 1$ (with $X \preceq Y$ meaning $x^{\mathsf{T}} X x \le x^{\mathsf{T}} Y x$ for every x): the displacement terms sum to $\eta^2 - \nu \ge \eta^2 - 1$, so the rms displacement beyond the noise is at least $(\eta^2 - 1)^{1/2}$ error bars.

The response is linear and both the noise and the residual are Gaussian, so the pull $\zeta = (\hat{p} - p_{\rm true})/\sigma_p$ is Gaussian with mean μ and variance $V = \langle \chi^2_p \rangle - \mu^2$, and the 1σ bar covers the truth in a fraction

$$P(|\zeta| < 1) \;=\; \Phi\!\left(\frac{1-\mu}{\sqrt{V}}\right) - \Phi\!\left(\frac{-1-\mu}{\sqrt{V}}\right) \tag{62}$$

of the universes, with $\Phi(x) = \frac{1}{2}\big[1 + \mathrm{erf}(x/\sqrt{2})\big]$. When the prior absorbs the known drift ($\mu \approx 0$),

$$P(|\zeta| < 1) \;=\; 2\,\Phi(1/\eta) - 1\,: \tag{63}$$

68% at $\eta = 1$, 50% at 1.5, 38% at 2. For $\eta \ge 1$ a nonzero μ only lowers the coverage, so Eq. (63) is then an upper bound. For the pessimistic and free priors it agrees with Eq. (62) within about 0.1 percentage points; for the growth-tied prior only Eq. (62) is right.

Under the known drifts plus the pessimistic scatter and the fiducial model, at $k_{\rm max} = 0.2\,h/\mathrm{Mpc}$ on $\Delta d_{\rm com} = 62$ Mpc, $\Delta z = 0.02$, 0.05 and 0.2, the growth-tied prior's Ω_m bar covers the truth in 6.3, 2.5, 10 and 35% of universes, where Eq. (63) would give 23, 19, 25 and 42%; the pessimistic prior covers in 67–68%, and the free prior in 68.3% exactly. For (w_0, w_a) the coverage depends on how the excess of $\langle \chi^2_w \rangle$ splits between the two directions, so only η is quoted.

A prior narrower than the truth. A prior narrower than the truth's scatter charges a large penalty for bin biases that follow the truth's wiggles, so the fit follows them only in part and moves the cosmology to absorb the rest, differently for every truth; the growth-tied prior,

with no residual freedom at all, is the extreme case. At $\eta = 2$ the real scatter of the best-fit (w_0, w_a) is twice as wide per axis as the reported ellipse, and the FoM four times too large. The opposite mismatch gives η below 1: a conservative ellipse.

The matched case. Collect all parameters in θ, with every prior centered on the fiducial, and let $\Delta\theta$ be the truth's departure from it: zero in the cosmology and in every parameter with another Gaussian prior, $B_g\,\delta_i$ in the amplitude β_i. With $\mathbf{n}_d$ the data noise, of covariance C_d, and $F = \mathsf{D}^\mathsf{T}\mathsf{C}_d^{-1}\mathsf{D} + F_\mathrm{bias} + F_\mathrm{other}$, where F_bias is the bias prior's precision, the linear response gives the fit's error

$$\hat{\theta} - \theta_\mathrm{true} \;=\; F^{-1}\big(\mathsf{D}^\mathsf{T}\mathsf{C}_d^{-1}\mathbf{n}_d \;-\; F_\mathrm{bias}\,\Delta\theta\big)\,: \tag{64}$$

the noise, of covariance $F^{-1}F_0F^{-1}$, plus the part of the truth's departure that the bias prior held back. In the matched case P_mem is built from the truth's kernel Σ, and every projection of that form obeys

$$\mathsf{P}_\mathrm{mem}\,\Sigma\,\mathsf{P}_\mathrm{mem} \;=\; \mathsf{P}_\mathrm{mem}\,: \tag{65}$$

with $\mathsf{Q} = \Sigma^{-1}$ and $\mathsf{M} = (\mathsf{U}^\mathsf{T}\mathsf{Q}\mathsf{U})^{-1}$, $\Sigma\,\mathsf{P}_\mathrm{mem} = \mathsf{I} - \mathsf{U}\mathsf{M}\mathsf{U}^\mathsf{T}\mathsf{Q}$, and $\mathsf{P}_\mathrm{mem}\mathsf{U} = 0$ removes the second term. The scatter part of Eq. (64) then has covariance $F^{-1}F_\mathrm{bias}F^{-1}$, the two covariances add up to F^{-1}, and the scatter and noise terms of Eq. (60) sum to exactly 2: the widening the prior adds to the ellipse equals the error the scatter inflicts. The matched value is $\eta = (1+\Delta^2_\mathrm{drift}/2)^{1/2}$ for (w_0, w_a) and $(1 + \Delta^2_{\mathrm{drift},p})^{1/2}$ for one parameter; for the fiducial model on the pessimistic prior, whose tilts absorb the drifts' lines, the measured value stays below 1.03 in every score, at every cut and on every layout (Section 6).

When a fit is conservative. Write P_mem in its null-space form, Eq. (52), with the fit's kernel Σ_fit, and let Σ_true be the truth's. If $\mathsf{N}^\mathsf{T}\Sigma_\mathrm{true}\mathsf{N} \preceq \mathsf{N}^\mathsf{T}\Sigma_\mathrm{fit}\mathsf{N}$, then $\mathsf{P}_\mathrm{mem}\Sigma_\mathrm{true}\mathsf{P}_\mathrm{mem} \preceq \mathsf{P}_\mathrm{mem}$ and $\eta \le (1 + \Delta^2_\mathrm{drift}/2)^{1/2}$: a prior wider than the truth wastes information but does not understate the error. The free prior has $F_\mathrm{bias} = 0$, so only the noise remains in Eq. (64) and $\eta = 1$ for every truth expressible through the bin biases.

K.5 The same test for Ω_m and A_s

Ω_m and A_s are scored the same way, each in units of its own marginalized error, with $\eta = \langle\chi^2_p\rangle^{1/2}$, the case $n_\mathrm{dof} = 1$ of Eq. (9). They can react more strongly to a bias error than (w_0, w_a), because a bias error moves the clustering amplitude $b_i^2 A_s D^2(\bar{z}_i)$ of bin i, which A_s scales directly and Ω_m moves through the growth factor.

L Numerical verification

The script `verify_prior.py`, in this folder, checks every identity of these appendices with the cocoa Python: 23 checks, each printing one PASS or FAIL line with the measured deviation. They cover the two-bin, three-bin, even and uneven tridiagonal inverses; the projection's null direction, semidefiniteness and rank; the Pythagorean split and the Schur complement; the neighbor-prediction leftover and, by Monte Carlo, the jump variance with the worked example of Appendix A.2; the continuity equation against Eq. (17), the conserved deviations and the neighbor rms values; and for the memory model the annihilation of U with rank $n - 2$, the $1/W^2$ Woodbury limit, the null-space identity, the three-bin curvature penalty, the vanishing

for one and two bins, the kriging predictor, the end-to-end correlation, the straight-line fits of Appendix C.4 and the LRG3 magnitude-limit rate. The theory figures are drawn from data written by `plots/make_theory_data.py`, with the same growth factor and fiducial amplitudes.

M Implementation and the Fisher construction

M.1 The model in the likelihood

The project likelihood (`likelihood/desi_cl.py`) has two bias options, `higher_order_bias`: 0 is linear bias, and 1 the one-loop model with every term per bin ($b_{1,i}$, Δ_2, Δ_s, α_0, α_2, SN_0, SN_2; $b_3 = 0$), in which a zero amplitude switches a term off at no cost. The fiducial model is one parameter file of option 1 per layout, which states the grouping, frees the tilts and fixes $\alpha_2 = 0$ in every bin (`scripts/make_desi_params.py --model-b`; "model b" in the script flag and the file names is the historical key of the fiducial widths; a fit without tilts holds the tilt parameters fixed); the paper yamls `EXAMPLE_EVALUATE1–4.yaml` run the four layouts, in layout order. A narrower one-loop variant built for $k_{\max} = 0.075\,h/\mathrm{Mpc}$, which `notes/higher_order_bias` records as the motivation for the fiducial widths, is not shipped. The likelihood computes $b_{2,i}$ and $b_{s^2,i}$ from $b_{1,i}$ with Eq. (5) and passes them to CosmoLike per bin.

- **Counterterms.** CosmoLike's k^2 slot is $2\,b_1 b_K\,k^2 P_{\mathrm{mm}}$, with b_K its per-bin derivative-bias coefficient and k in units of H_0/c, so the likelihood sets $b_K = \alpha_0/[2\,b_1(c/H_0)^2]$, with $c/H_0 = 2997.92\,h^{-1}\mathrm{Mpc}$. α_0 therefore multiplies the nonlinear matter spectrum, where DESI's definition multiplies the linear one (`notes/higher_order_bias`, subsection "One likelihood option, every term per bin"). α_2, in $h^{-2}\,\mathrm{Mpc}^2$, has its own setter and multiplies the linear spectrum, as in Eq. (6).

- **Stochastic terms.** In Eq. (7), $P_{ii} = 1/\bar{n}_i$ is fixed by the bin's surface density, so that the projection with the raw distribution gives the angular shot noise. A cross pair of two bins of one tracer sample takes the mean of their two powers; a pair of two samples takes none, because different galaxies share no stochastic term.

- **Projection.** The low-rank non-Limber engine projects each bin's full one-loop $P_{gg,i}$, counterterm included, and the adjacent cross pairs carry the bilinear one-loop pair spectrum (`notes/higher_order_bias`, subsection "Separability and the D^4 rule"). Appendix O shows why the counterterm requires this engine.

M.2 The derivatives

The derivatives of the data vector m (the band powers) come in three kinds, all computed on the paper yamls and cached by the notebook `EXAMPLE_EVALUATE_FISHER_MODEL.ipynb` (or, identically, by the MPI script `scripts/compute_fisher_base_mpi.py`). *Stencil derivatives*, five-point stencils of the full likelihood: the seven cosmological parameters ($10^9 A_s$, the spectral index n_s, the Hubble constant H_0, the baryon density Ω_b, Ω_m, w_0, w_a), the six group amplitudes B_g and the seven velocity dispersions. *Per-bin operator columns* $\partial m/\partial b_{1,i}$, $\partial m/\partial b_{2,i}$, $\partial m/\partial b_{s^2,i}$ and $\partial m/\partial \alpha_{0,i}$, each at fixed values of the others, from the same five-point stencil through the likelihood's setters (steps of 1% of b_1, 0.02 in b_2 and b_{s^2}, $5\,h^{-2}\,\mathrm{Mpc}^2$ in α_0). The bins shifted together form one color of a greedy coloring of the cross-pair graph, in which two bins joined by a measured cross pair never share a color, so each spectrum involves at most one shifted

bin; every layout needs four colors. *Exact templates*: α_2, SN_0 and SN_2 are linear in their amplitudes, so their per-bin columns are exact (Eq. (7) for the stochastic terms).

The calibrations tie b_2 and b_{s^2} to b_1, so the derivative with respect to a bin's linear bias is the chain-rule sum

$$\frac{dm}{db_{1,i}} \;=\; \frac{\partial m}{\partial b_{1,i}} \;+\; b_2^{\mathrm{L}\prime}(b_{1,i})\,\frac{\partial m}{\partial b_{2,i}} \;-\; \frac{4}{7}\,\frac{\partial m}{\partial b_{s^2,i}}, \tag{66}$$

with $b_2^{\mathrm{L}\prime}$ the slope of the Lazeyras fit. Summed over a group with weights $1/D(\bar{z}_i)$, these columns must reproduce the stencil derivative of the amplitude B_g (Appendix N).

M.3 The variants and the Fisher matrices

The parameters are the cosmology and the velocity dispersions (with the Gaussian priors of Table 2), the variant's higher-order parameters, and one amplitude $\beta_i = b_{1,i}\,D(\bar{z}_i)$ per bin (Eq. (12)), whose column is Eq. (66) divided by $D(\bar{z}_i)$. The higher-order columns of group g are sums over its bins,

$$\frac{\partial m}{\partial \Delta_{2,g}} = \sum_{i\in g} \frac{\partial m}{\partial b_{2,i}}, \qquad \frac{\partial m}{\partial t_{2,g}} = \sum_{i\in g} (\bar{z}_i - \bar{z}_g)\,\frac{\partial m}{\partial b_{2,i}}, \tag{67}$$

and likewise for the other terms, with $t_{2,g}$ the tilt of Δ_2 in group g. The fit with tilts adds the tilt columns of Δ_2, Δ_s, α_0 and α_2, with flat priors, in every group of more than one bin: the free straight line in redshift that s_g gives the linear bias.

The data Fisher matrix is $F_{\mathrm{data}} = \mathsf{D}^{\mathsf{T}}\mathsf{C}_d^{-1}\mathsf{D}$ on the kept entries, with D the matrix of the columns and C_d the covariance of Section 3; the Gaussian priors add to its diagonal, and the linear-bias prior adds its projected precision to the amplitude block (Appendix J). In the $k_{\max}$ scan each variant's derivatives are a fixed combination $\mathsf{D} = \mathsf{B}\mathsf{M}$ of the base columns B, so $F_{\mathrm{data}} = \mathsf{M}^{\mathsf{T}}(\mathsf{B}^{\mathsf{T}}\mathsf{C}_d^{-1}\mathsf{B})\,\mathsf{M}$ on the entries each cut keeps: 168 systems (four layouts, seven cuts, two variants, three priors), 162 with a fit and 6 without any band. Every Fisher matrix is inverted through a pseudo-inverse, an inverse taken only over the directions that carry information (Appendix N).

M.4 The priors, the response and the truths

- **Linear-bias priors.** Growth-tied sums the amplitude columns of each group into one; the baseline and pessimistic priors add the projected precision of Eq. (24) to the amplitude block; free adds nothing.

- **Known drifts and random truths.** The two drift laws of Appendix K.2 plus residuals with the pessimistic kernel; a drift δ_i of the linear bias gives $\delta m = \sum_i (\partial m/\partial\beta_i)\,\beta_i\,\delta_i$, with the column of Eq. (66), the fit moves by Eq. (56), and the average over the residuals is exact (Appendix K.3).

- **Calibration drift.** One deterministic truth that moves only the offsets, $\delta m = \sum_i [(\partial m/\partial b_{2,i})\,\Delta_{2,i} + (\partial m/\partial b_{s^2,i})\,\Delta_{s,i}]$, with $\Delta_{2,i} = (\bar{z}_i - \bar{z}_g)/(\bar{z}_{\max} - \bar{z}_{\min})$ and $\Delta_{s,i} = \Delta_{2,i}/2$, where $\bar{z}_{\max}$ and $\bar{z}_{\min}$ are the largest and smallest mean redshifts of the group's bins. It changes by 1 in Δ_2 and 0.5 in Δ_s across the group and has zero mean over its bins; a one-bin group gets zero.

- **Break truths.** δm is the α_2 or SN_2 column times the truth's value, or the b_2 and b_{s^2} columns times the offsets $\pm(2, 1)$, summed over every bin, or the calibration drift above.

Each figure has its own folder under `plots/`, with its data (`data.json`), the script that draws it from that file alone, and the PDF. The data come from the notebook `EXAMPLE_EVALUATE_FISHER_MODEL.ipynb`.

N Numerical checks

- **The chain rule.** For every group, the per-bin columns of Eq. (66), summed with the weights $1/D(\bar{z}_i)$, reproduce the stencil derivative of the group amplitude B_g to at most 1.5×10^{-9} of the stencil column's largest entry on every layout, the truncation error of the two stencils. The two routes are independent: the operator columns go through the bias setter, the stencil through the full likelihood.

- **The templates.** The template cache's order-0 templates equal the derivative cache's SN_0 templates, and the two caches' fiducial vectors have $\Delta\chi^2 = 0$ on every layout.

- **The α_2 columns.** The forward and backward differences agree to at most 1.6×10^{-10} of each column (4.3×10^{-11} on the three thinner layouts), and the vector after the steps equals the one before bit for bit: the term is linear, and the setter was reset.

- **Linear response.** The response is linear in the truth, so the + and − truths of the shift estimates must give equal scores: the largest difference is 0.

- **Directions without information.** At small cuts a flat-prior tilt whose group keeps few bins (on $\Delta d_{\mathrm{com}} = 62$ Mpc at $k_{\mathrm{max}} = 0.075\,h/\mathrm{Mpc}$ only 7 of the 18 BGS bins keep any band) can leave one combination of parameters with no information, which a plain inverse amplifies into a spurious shift. The notebook therefore scales every Fisher matrix to unit diagonal, diagonalizes it, and drops the directions below 10^{-12} of the largest eigenvalue: 11 of the 168 systems of the scan drop one, with a largest component on (w_0, w_a), Ω_m and A_s of 3.3×10^{-10}, pure nuisance combinations.

- **The σ_v priors.** The widths stored in the derivative caches (the paper yamls') are exactly six times the exploration widths of Table 2 that this note's α_2-free fits use.

- **The loop templates.** With $b_3 = 0$, CosmoLike's C-FAST-PT loop templates and Python FAST-PT agree on the $\Delta d_{\mathrm{com}} = 62$ Mpc vector, pairs included, to $\Delta\chi^2 \leq 3.5 \times 10^{-5}$ (`notes/higher_order_bias`, section "Validation").

O The counterterm requires the low-rank engine

CosmoLike has two non-Limber engines for C_ℓ^{gg}. FKEM, the method of Fang et al. [8] built on FFTLog (a fast Hankel transform on a logarithmic grid), projects the linear term exactly and moves everything nonlinear, the counterterm included, through its Limber difference. The low-rank engine factorizes each bin's full one-loop P_{gg}, less the white-noise constant of the loops (which Limber projects exactly), and projects it without the Limber approximation.

On $\Delta z = 0.05$ the low-rank response $\partial C_\ell/\partial\alpha_0$ is 7.3 times the Limber one at $\ell = 20$, 1.8 times at $\ell = 50$, 1.26 times at $\ell = 100$ and about 1.05 times at $\ell = 200$–400 (median over the kept autos). Limber evaluates the power at the transverse wavenumber $k = (\ell + 1/2)/\chi$ only, while a thin bin with sharp edges also passes radial modes of higher k, which the k^2 weight amplifies. The low-rank response is converged: raising `nonlimber_accuracyboost` from 32 to 128, the rank from 5 to 10, or the overall accuracy boost from 1 to 2 changes it by a $\Delta\chi^2$ far below one. For a one-sigma α_0 the two engines' responses differ by tens of units of χ^2.

P Limitations

- **Linear response.** The shifts and $\sqrt{\langle\chi^2\rangle/n_{\rm dof}}$ are first order in the truth's departure from the fiducial. The dropped second-order term is a few percent of the shift for 1–3% residuals and perhaps 10–20% for the 22% BGS drift (Appendix K.3).

- **Theory truths, not mocks.** Residuals rougher than the pessimistic prior, and calibrations that drift nonlinearly within a group, are untested.

Q Open issues

- **One-loop terms of the covariance in Limber.** The covariance carries the one-loop terms of the shared fiducial (b_2 and b_{s^2} on their calibrations) through the Limber approximation, and its low-ℓ non-Limber recompute is linear. In the lowest bands of thin bins that leaves an error of at most 1.3% of C_ℓ.

- **The cut is computed on the fiducial spectra.** The UV-fraction cut uses the paper model's fiducial vector, not the model's response templates. On $\Delta z = 0.05$ at $k_{\rm max} = 0.075\,h/{\rm Mpc}$ the UV fraction of Eq. (1), applied to the α_0 template $\partial C_\ell/\partial\alpha_0$, is a median 0.076 and a maximum 0.14 on the kept bands, against 0.017 and 0.052 for the fiducial one-loop vector, and about half of each bin's SN_2 auto kernel comes from line-of-sight wavenumbers above the bin's own cut: part of what the forecast learns about α_0 and SN_2 comes from scales the cut is meant to exclude.

- **Negative α_0 in chains.** At $\alpha_0 = -88.9\,h^{-2}\,{\rm Mpc}^2$, one $k_{\rm max} = 0.075\,h/{\rm Mpc}$ prior width below zero, the spectrum the low-rank engine factorizes, P_{gg} less its white-noise constant, turns negative above $k \approx 0.14\,h/{\rm Mpc}$; the engine floors 47% of its grid, and the vector curves in α_0. The Fisher forecast at $\alpha_0 = 0$ does not see this, but a chain that explores negative α_0 does. Neither remedy, a loop taper or a positivity prior on P_{gg} (`notes/higher_order_bias`, subsection "The white-noise limit of the renormalized loops"), is implemented.

- **The b_3 template is wrong.** CosmoLike's third-order bias template (`tab_d1d3` of C-FAST-PT) is a fixed, cosmology-independent table that disagrees with Python FAST-PT, so the model sets $b_3 = 0$, as DESI's baseline does.

- **No redshift-space one-loop terms.** The model adds real-space loops to linear Kaiser distortions and relies on the sampled σ_v and on α_2 for the rest.

- **Adjacent cross spectra see the shared edge.** The cross spectra of adjacent thin bins are sensitive to small-scale radial physics at their common edge, as SN_2 shows; the model carries SN_2 for that reason, and any other radial physics on that scale reaches the cross spectra the same way.

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